

REQUIREMENTS / EVIDENCE / INTERFACES / DECISIONS

Engineering English

Challenge the assumption. Explain the evidence.



Phil Stilwell

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Eight lessons. Eleven extended conversations.

INSIDE THE BOOK

192 vocabulary entries / 128 reusable phrases
220 extended-dialogue turns / 11 transfer exchanges
Word banks and explained answer keys

Upper-intermediate to advanced (B2-C1). For design, systems, test, reliability, manufacturing, and field engineers.

2026 edition

Eight scenarios. Eight skills.

Eight technical conversations that connect requirements, engineering evidence, trade-offs, and decisions.

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Precision makes disagreement productive.

Engineering conversations move between customer needs, models, drawings, measurements, and decisions. A familiar word such as verified, robust, or compatible can mean different things to different teams. These cases develop the language for naming the exact claim, testing its basis, and explaining what remains unresolved without making the discussion personal.

Name the reference

A result needs a requirement, revision, condition, or comparison point. Identify the reference before saying the design passes, improves, or remains compatible.

"The result applies to revision B under the recorded test conditions."

Separate observation from explanation

A failed response is an observation. A suspected mechanism is an explanation to investigate. Preserve the distinction even when the hypothesis sounds convincing.

"The output dropped during the test; connector damage is not yet confirmed."

Make the trade-off explicit

Lower material cost can accompany higher assembly cost. A stronger part may add mass. State the comparison basis and the constraints instead of presenting one favorable number as the whole decision.

"Option B costs less at the stated volume after tooling is included."

Close with evidence and ownership

A meeting can identify a gap without closing it. Name who will supply the missing evidence, who reviews it, and what decision remains pending.

"Please attach the interface analysis before the release owner reviews closure."

PROFESSIONAL BOUNDARIES

Original fictional English practice, not engineering design guidance, safety certification, professional approval, or instructions for operating equipment. Numbers and technical cases are invented. Actual design, testing, field action, and compliance decisions require qualified personnel, applicable requirements, approved procedures, and the relevant authority.

Lightweight compared with what?

A Read the situation

Convert an ambiguous customer request into a measurable proposal while keeping unresolved conditions and approval visible.

Customer representative Julia asks engineer Wei for a lightweight, durable enclosure. Their meeting proposes a maximum mass of 1.2 kg for the empty enclosure including its cover and fasteners. Electronics are excluded. The proposed service life is five years, but operating temperature, humidity, handling conditions, and the verification method are unresolved. Nothing is approved. Wei must record the proposed mass limit accurately and avoid turning the life target into a demonstrated durability claim.

PEOPLE / Julia: Customer representative | Wei: Design engineer

ASK FOR THE BOUNDARY BEHIND THE ADJECTIVE

A request for a robust or lightweight product communicates a preference, not a complete acceptance basis. Ask what is included, under which conditions, and how success will be checked. Keep the discussion focused on the product rather than criticizing the customer for imprecise language.

Briefing checks

Circle one answer in each item. Use the briefing.

1. What is included in the proposed mass limit?

- A Shipping packaging and spare parts
- C All installed electronics

- B The empty enclosure, cover, and fasteners
- D Only the removable cover

2. What is the status of five years?

- A A demonstrated test result
- C A measured failure-free period

- B An approved warranty promise
- D A proposed service-life target

3. Which condition is still open?

- A Operating temperature
- C The inclusion of fasteners

- B The proposed 1.2 kg mass value
- D The exclusion of electronics

[Answers and explanations: page 85.](#)

The vocabulary that does the work

B Find the words / 1 of 2

Read the definitions and say the common word combinations aloud.

stakeholder need

The capability or outcome a stakeholder wants.

IN USE capture the stakeholder need

design requirement

A specified condition the design must satisfy.

IN USE define a design requirement

design constraint

A limit on available design choices.

IN USE work within a design constraint

mass budget

The planned allocation of allowable mass.

IN USE allocate the mass budget

maximum mass

The greatest permitted mass under a stated boundary.

IN USE specify maximum mass

system boundary

The line defining what is included in the system.

IN USE define the system boundary

operating envelope

The range of conditions for intended operation.

IN USE define the operating envelope

service life

The intended period of use under specified conditions.

IN USE state the service life

duty cycle

The pattern of operation and rest over time.

IN USE specify the duty cycle

environmental condition

A surrounding condition affecting operation or storage.

IN USE document environmental conditions

acceptance criterion

A condition used to judge whether a result is acceptable.

IN USE agree an acceptance criterion

verification method

The means used to check a stated requirement.

IN USE select a verification method

Precision check

Mass is not the same as the mass of the fully populated product. The proposal includes the enclosure, cover, and fasteners but excludes electronics. A five-year target needs defined use conditions and a suitable evidence plan before it can support a durability claim.

The vocabulary that does the work

B Find the words / 2 of 2

Read the definitions and say the common word combinations aloud.

testability

The extent to which a claim can be checked by testing.

IN USE improve testability

requirement allocation

Assignment of a requirement to a system element.

IN USE review requirement allocation

derived requirement

A requirement developed from higher-level needs or analysis.

IN USE justify a derived requirement

design margin

Allowance between a predicted value and a defined limit.

IN USE preserve design margin

threshold

A stated boundary for an acceptable result.

IN USE specify the threshold

objective value

A desired value beyond a minimum acceptable level.

IN USE distinguish threshold and objective

assumption register

A record of unverified premises used in planning or design.

IN USE update the assumption register

traceability

Recorded connections between needs, requirements, and evidence.

IN USE maintain traceability

baseline

An approved reference placed under change control.

IN USE establish a baseline

ambiguity

Wording that permits more than one interpretation.

IN USE resolve ambiguity

rationale

The reason supporting a requirement or decision.

IN USE record the rationale

requirement owner

The role responsible for maintaining a requirement.

IN USE identify the requirement owner

Precision check

A proposed value is not an approved baseline. Separate what the meeting clarified from what remains open. Terms such as shall, should, and target also need the meaning assigned by the governing specification, not an assumed universal interpretation.

Phrases you can actually use

C Notice the language / 1 of 2

CLARIFY THE ADJECTIVE

"What maximum mass would count as lightweight?"

EXCLUDE EXPLICITLY

"Electronics are outside this mass boundary."

QUALIFY THE STATUS

"This is a proposal, not an approved requirement."

CLARIFY THE LIFE CLAIM

"Five years under which use and maintenance conditions?"

SET THE BOUNDARY

"Does that value include the cover and fasteners?"

STATE THE PROPOSAL

"We propose a maximum mass of 1.2 kg."

ASK ABOUT CONDITIONS

"What operating temperature range must the enclosure withstand?"

IDENTIFY A MISSING INPUT

"The handling conditions have not been specified."

Lightweight

Needs a value, boundary, and comparison context.

Durable

Needs relevant conditions and a defined life or performance basis.

Maximum

A limit, not the same as a preferred nominal value.

Phrases you can actually use

C Notice the language / 2 of 2

SEPARATE TARGET FROM EVIDENCE

"The service-life target is not a demonstrated result."

KEEP THE REQUIREMENT SINGULAR

"Separate the mass limit from the service-life requirement."

CONNECT THE NEED

"Which customer task makes this limit necessary?"

REQUEST OWNERSHIP

"Who will confirm the environmental requirements?"

ASK HOW TO CHECK

"Which verification method will support acceptance?"

PROTECT THE BOUNDARY

"A packaging change should not silently alter the measured assembly."

RECORD A PREMISE

"Put the unconfirmed condition in the assumption register."

CLOSE ACCURATELY

"We have clarified the proposal, but approval is still pending."

Including

Identifies items counted in the stated boundary.

Proposed

Keeps a discussion value separate from approval.

Demonstrated

Requires evidence appropriate to the exact claim.

Choose the precise message

D Check your understanding

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

1. Which statement accurately records the mass proposal?

- A** Approximately 1.2 kg for any version of the complete product
- B** A guaranteed 1.2 kg for packaging and all components
- C** An approved mass limit for electronics alone
- D** Maximum 1.2 kg for the empty enclosure including cover and fasteners, excluding electronics

2. Which sentence avoids overstating durability?

- A** Five years is the proposed service life; operating conditions and verification remain open.
- B** The enclosure has already survived five years.
- C** The proposal proves performance in every environment.
- D** A mass limit establishes service life automatically.

3. What should happen to unresolved environmental assumptions?

- A** Convert each into a favorable test result.
- B** Let every supplier choose a different interpretation without recording it.
- C** Record them with owners for confirmation.
- D** Remove them to make the requirement look finished.

4. A prototype meets the proposed mass boundary at 1.15 kg. What does that establish about the five-year target?

- A** The operating envelope is automatically approved.
- B** The electronics must be included because the measured mass is below the limit.
- C** Nothing by itself; a mass result does not demonstrate service life under undefined conditions.
- D** Five-year life is verified because 1.15 is below 1.2.

Answers and explanations: [page 85](#).

Lightweight compared with what?

E Conversations / turns 1-10

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

WORD + PHRASE BANK

baseline / assumption register / traceability / service life / verification method / requirement owner / rationale / system boundary / maximum mass / operating envelope

- 01 **Julia:** We need the new enclosure to be lightweight and durable. The previous proposal felt too heavy, but our brief does not yet give a numerical limit.
- 02 **Wei:** Let us define the ¹ _____. Would 1.2 kilograms be the proposed upper limit, and what exactly should be included when we measure it?
- 03 **Julia:** Use that value for discussion. I mean the empty enclosure with its cover and fasteners, not the electronics that our team will install later.
- 04 **Wei:** Then the ² _____ includes those three items and excludes electronics. I will put that beside the number so nobody compares it with a populated assembly.
- 05 **Julia:** For durability, our commercial team has suggested five years. They have not told us the expected operating temperature or how often the unit will be handled.
- 06 **Wei:** Five years is a proposed ³ _____, not yet a demonstrated result. We need the use conditions before selecting evidence that could support the requirement.
- 07 **Julia:** Could we leave the conditions to the design team? I do not want to slow the project by asking the customer for every detail immediately.
- 08 **Wei:** We can propose an ⁴ _____, but the customer needs to confirm it. Otherwise we might optimize for conditions that do not match the actual installation.
- 09 **Julia:** I will ask about temperature, humidity, and handling. Some units may move between workstations, so we should not assume they all stay fixed in one installation.
- 10 **Wei:** Record each unresolved premise in the ⁵ _____. Give it an owner and a confirmation point rather than hiding it inside a confident statement about durability.

The conversation, continued

E Conversations / turns 11-20

Continue with the word bank on page 10.

- 11 **Julia:** Once the conditions are clear, I assume we will need a way to show that the proposal has been met, not just a convincing description.
- 12 **Wei:** Yes. The ⁶ _____ must suit the requirement. We should agree what is checked, on which configuration, and what evidence the acceptance decision will use.
- 13 **Julia:** I also want the specification to preserve why we chose the limit. A later reviewer may otherwise change it without understanding the installation problem behind it.
- 14 **Wei:** Keep the ⁷ _____ with the requirement. That helps reviewers assess whether a proposed change still serves the customer need instead of debating the number in isolation.
- 15 **Julia:** Let us send the precise mass proposal now and mark the environmental questions as open. I do not want the clear mass limit to make the whole specification look settled.
- 16 **Wei:** Correct. Clear ⁸ _____ will connect the need, the agreed wording, and the eventual evidence. It also makes the unresolved links visible during review.
- 17 **Julia:** I will send the customer the proposed boundary and the missing questions. Please do not label the specification approved until their review is complete.
- 18 **Wei:** We will establish the ⁹ _____ only through the agreed approval process. Today's useful result is a precise proposal and an explicit list of remaining decisions.
- 19 **Julia:** Good. I can explain that progress without claiming the enclosure has already met the five-year target. Who should keep the environmental questions moving?
- 20 **Wei:** Name a ¹⁰ _____ for that follow-up. The record should say who confirms the conditions and when, so the open questions do not disappear between teams.

ANSWERS / EXPLANATIONS: PAGE 86

1. maximum mass | 2. system boundary | 3. service life | 4. operating envelope | 5. assumption register | 6. verification method | 7. rationale | 8. traceability | 9. baseline | 10. requirement owner

A boundary changes the meaning

F Say it

A proposal limits an empty housing with lid and screws to 0.8 kg. Batteries are excluded. The operating humidity and acceptance test are not yet agreed. Approval remains pending.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

unresolved / included / excluded / baseline

Engineer: "The lid and screws are ¹ _____."

Customer: "The batteries are ² _____."

Engineer: "Humidity remains ³ _____."

Customer: "This is a proposal, not an approved ⁴ _____."

ANSWERS / EXPLANATIONS: PAGE 86

1. included | 2. excluded | 3. unresolved | 4. baseline

Rehearse with a fixed script

1. Read the requirements dialogue. Stress the enclosure, cover, and fasteners included in 1.2 kg, with electronics excluded.
2. Switch roles. Keep five years as proposed service life while operating conditions, verification, and approval remain open.
3. Complete and check the transfer. State the 0.8 kg boundary without adding batteries or an approved humidity condition.

WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

The drawing fits; the evidence is missing

A Read the situation

Challenge a release recommendation by identifying a specific evidence gap and proposing a bounded closure action.

At a design review, mechanical lead Ahmed and reviewer Clara examine an enclosure interface. The CAD model shows 2 mm nominal clearance. Requirement R-14 specifies at least 1 mm clearance across the approved dimensional tolerances. The tolerance stack-up has not been documented, and the supplier drawing revision used in the model is unconfirmed. The chair is preparing to recommend release. Clara must distinguish nominal fit from verified minimum clearance and record a specific closure action without claiming the design has already failed.

PEOPLE / Ahmed: Mechanical lead | Clara: Design reviewer

PUSH BACK ON THE CLAIM, NOT THE PERSON

A direct technical challenge can be cooperative when it identifies the requirement, the missing evidence, and the decision affected. State what would resolve the concern. Avoid either approving from politeness or treating an incomplete analysis as proof that a colleague designed a defective product.

Briefing checks

Circle one answer in each item. Use the briefing.

1. What does the CAD model currently show?

- A The approved supplier revision
- C 2 mm nominal clearance

- B A completed tolerance analysis
- D A verified 1 mm minimum across all tolerances

2. What does R-14 require?

- A At least 1 mm across approved dimensional tolerances
- C A nominal value greater than zero only

- B Exactly 2 mm in every assembly
- D An undocumented supplier agreement

3. What is not established?

- A Whether the minimum clearance meets R-14
- C Whether R-14 contains a numerical limit

- B Whether the CAD view shows a nominal gap
- D Whether a release recommendation is being prepared

Answers and explanations: page 87.

The vocabulary that does the work

B Find the words / 1 of 2

Read the definitions and say the common word combinations aloud.

design review

A structured examination of a design and its supporting evidence.

IN USE conduct a design review

review package

The drawings, analyses, and records provided for review.

IN USE complete the review package

nominal dimension

The stated reference dimension before permitted variation.

IN USE check the nominal dimension

clearance

The space between adjacent parts.

IN USE verify minimum clearance

interference

Unwanted overlap or contact between parts.

IN USE identify potential interference

dimensional tolerance

The permitted variation in a specified dimension.

IN USE apply dimensional tolerances

tolerance stack-up

The combined dimensional variation affecting an assembly condition.

IN USE calculate the tolerance stack-up

worst-case analysis

An assessment using the most adverse allowed combination.

IN USE document a worst-case analysis

datum

A reference used to establish dimensional relationships.

IN USE identify the datum

mating surface

A surface that joins or interfaces with another part.

IN USE inspect the mating surface

CAD model

A computer-aided design representation of geometry.

IN USE update the CAD model

drawing revision

The identified version of an engineering drawing.

IN USE confirm the drawing revision

Precision check

A nominal 2 mm gap does not establish the minimum gap across permitted variation. Missing evidence also does not prove physical interference. The supported finding is that R-14 is not yet verified on a confirmed configuration.

The vocabulary that does the work

B Find the words / 2 of 2

Read the definitions and say the common word combinations aloud.

assumption log

A record of premises used in an analysis.

IN USE review the assumption log

open item

An unresolved matter recorded for follow-up.

IN USE assign an open item

closure evidence

Information demonstrating that an open item is resolved.

IN USE attach closure evidence

review action

A task assigned as a result of a review.

IN USE record a review action

technical objection

A reasoned challenge to a technical claim or decision.

IN USE state a technical objection

dissent

A recorded difference of professional judgment.

IN USE document unresolved dissent

release recommendation

Advice that a design proceed to a release decision.

IN USE qualify a release recommendation

release authority

The role authorized to approve the relevant release.

IN USE identify the release authority

conditional approval

Approval explicitly subject to specified conditions, where allowed.

IN USE document conditional approval

concurrence

Agreement with a stated position or disposition.

IN USE obtain reviewer concurrence

design freeze

A defined point after which design changes require controls.

IN USE prepare for design freeze

action closure

Formal resolution of an assigned review task.

IN USE verify action closure

Precision check

Conditional approval is meaningful only if the local process permits it and records its conditions, restrictions, and authority. An informal promise to finish the analysis later must not be written as unconditional release approval.

Phrases you can actually use

C Notice the language / 1 of 2

LOCATE THE CONCERN

"My concern is the evidence for R-14."

DISTINGUISH THE CLAIM

"Nominal clearance is not minimum clearance across tolerances."

CHECK THE REFERENCE

"Which supplier drawing revision does the model use?"

STATE THE GAP

"We cannot yet verify the required minimum."

ACKNOWLEDGE THE WORK

"The nominal geometry is clear in the model."

ASK FOR THE ANALYSIS

"Where is the documented tolerance stack-up?"

AVOID ALLEGING FAILURE

"I am not saying interference has been demonstrated."

BOUND THE ACTION

"Confirm the revision and attach the stack-up for this interface."

Fits

May describe nominal geometry without addressing variation.

Minimum

Must be evaluated against the stated range of conditions.

Unverified

Does not mean demonstrated failure.

Phrases you can actually use

C Notice the language / 2 of 2

DEFINE CLOSURE

"Closure needs a reviewed result against R-14."

KEEP THE DECISION VISIBLE

"The release recommendation remains unresolved."

CLARIFY CONDITIONS

"What restrictions would apply to conditional approval?"

PREVENT SILENT CLOSURE

"Do not close the item because the meeting has ended."

NAME RESPONSIBILITY

"Ahmed will prepare the analysis for review."

SEPARATE MEETING ROLES

"Reviewer concurrence is not automatically release authority."

PRESERVE DISAGREEMENT

"Record the technical objection with its evidence basis."

INVITE RESOLUTION

"What evidence would let us resolve this concern today?"

No objection

Should not conceal an unresolved technical concern.

Conditional

Needs explicit conditions and an authorized process.

Closed

Requires the specified evidence, not merely an assigned task.

Choose the precise message

D Check your understanding

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

1. Which review comment is best supported?

- A** The design definitely interferes because no stack-up is attached.
- B** The nominal gap alone verifies every allowed assembly.
- C** The supplier revision cannot affect interface review.
- D** R-14 remains unverified pending the stack-up and revision confirmation.

2. Which closure action addresses both missing inputs for R-14?

- A** Confirm the supplier drawing revision and review the interface tolerance stack-up against the 1 mm minimum.
- B** Measure the nominal CAD gap again and mark the minimum verified.
- C** Confirm the drawing filename without checking the tolerances used.
- D** Obtain meeting attendance signatures and treat them as clearance evidence.

3. What may the minutes record now?

- A** A measured clearance failure without measurement or analysis
- B** Supplier concurrence that has not been received
- C** An open verification item and its assigned follow-up
- D** Unconditional release approval inferred from meeting attendance

4. What would justify closing the item?

- A** Repeating the 2 mm nominal value more confidently
- B** A later meeting date without new evidence
- C** A statement that similar designs usually fit
- D** Reviewed evidence for the confirmed configuration against R-14

Answers and explanations: page 87.

The drawing fits; the evidence is missing

E Conversations / turns 1-10

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

WORD + PHRASE BANK

Nominal clearance / technical objection / Conditional approval / drawing revision / review action / closure evidence / tolerance stack-up / open item / Action closure / release authority

- 01 **Ahmed:** The model shows a two-millimeter gap, and the chair wants to recommend release today. Do you have a specific concern, or can we close the review?
- 02 **Clara:** My concern is the ¹ _____. R-14 applies across the permitted dimensional variation, and the supporting calculation is not in the review package.
- 03 **Ahmed:** The nominal geometry is comfortably above one millimeter. I agree that it does not show every combination, but I do not want the team to hear that the design has failed.
- 04 **Clara:** Nor do I. ² _____ is two millimeters. The minimum remains unverified, which is different from saying that interference has been observed or established.
- 05 **Ahmed:** That is a fair distinction. The model also came from the supplier file, but I cannot confirm which drawing issue was used without checking our records.
- 06 **Clara:** Confirm the ³ _____ before completing the calculation. An analysis of an obsolete interface could look convincing while answering the wrong configuration question.
- 07 **Ahmed:** Please replace check fit with the actual task: confirm the supplier revision and assess minimum clearance against R-14. I can own that analysis.
- 08 **Clara:** Exactly. Make it a ⁴ _____ with an owner and a review point. The task is narrow enough to complete without reopening unrelated parts of the design.
- 09 **Ahmed:** I can own the analysis. I will identify the confirmed supplier revision, state the tolerance assumptions, and present the resulting minimum against R-14.
- 10 **Clara:** That provides the intended ⁵ _____. The reviewer still needs to assess the result; attaching a file is not itself the same as resolving the concern.

The conversation, continued

E Conversations / turns 11-20

Continue with the word bank on page 19.

- 11 **Ahmed:** Could the chair record conditional approval and allow the release team to continue? I want to avoid delaying work that does not depend on this interface.
- 12 **Clara:** ⁶ _____ is only available if our process permits it. The record would need clear restrictions and the appropriate decision maker, not a casual promise to finish later.
- 13 **Ahmed:** I will ask the chair to separate any permitted preparatory work from release. We should not imply that every downstream activity has the same dependency.
- 14 **Clara:** Agreed. Identify the ⁷ _____ and let that role decide on the recorded basis. My review comment should not be mistaken for unilateral approval or rejection.
- 15 **Ahmed:** That is a concern I can answer. Keep the comment tied to the missing stack-up and revision, so the team knows exactly what needs closing.
- 16 **Clara:** A ⁸ _____ should be answerable. Here the requirement, missing analysis, and configuration question make the concern concrete enough for us to resolve.
- 17 **Ahmed:** I will revise the minutes to keep the interface item open. The release recommendation will remain unresolved until the evidence and required review are available.
- 18 **Clara:** Keep that ⁹ _____ visible in the action list. A meeting ending or an owner accepting the task must not make its status appear complete.
- 19 **Ahmed:** Once the reviewed analysis is available, we can record the result and route the recommendation to the authorized decision maker with any remaining conditions.
- 20 **Clara:** Yes. ¹⁰ _____ should identify the evidence and reviewer. That gives the release decision a traceable basis while preserving what the team actually established.

ANSWERS / EXPLANATIONS: PAGE 88

1. tolerance stack-up | 2. Nominal clearance | 3. drawing revision | 4. review action | 5. closure evidence | 6. Conditional approval | 7. release authority | 8. technical objection | 9. open item | 10. Action closure

A nominal fit is not a verified minimum

F Say it

A model shows 3 mm nominal clearance. The requirement specifies at least 1 mm across tolerances. The stack-up is missing, and the supplier revision is not confirmed.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

nominal / evidence / unverified / confirmation

Reviewer: "Three millimeters is the ¹ _____ clearance."

Engineer: "The supplier revision needs ² _____."

Reviewer: "The minimum remains ³ _____."

Engineer: "The item closes with reviewed ⁴ _____."

ANSWERS / EXPLANATIONS: PAGE 88

1. nominal | 2. confirmation | 3. unverified | 4. evidence

Rehearse with a fixed script

1. Read the design review. Contrast the 2 mm nominal gap with R-14's minimum of 1 mm across tolerances.
2. Switch roles. Request the confirmed supplier revision and reviewed stack-up without announcing interference or release approval.
3. Complete and check the transfer. Keep the 3 mm nominal value distinct from the unverified minimum and evidence needed for closure.

WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

One interruption, several possible explanations

A Read the situation

Report a failure observation precisely, distinguish mechanism from cause, and propose evidence-led investigation language.

During a planned 60-minute bench run, sample S7 stops producing an output at minute 46. Test engineer Grace and reliability engineer Pavel review the event. The failure has not been reproduced. The test setup has not been checked, and connector damage is only a hypothesis. The team has not established a population failure rate or a corrective action. Grace must preserve the result and test records while preventing a draft report from naming connector damage as the confirmed root cause.

PEOPLE / Grace: Test engineer | Pavel: Reliability engineer

A PLAUSIBLE EXPLANATION IS STILL A HYPOTHESIS

Technical teams often converge quickly on a familiar failure mechanism. Use language that invites testing of the explanation without dismissing experience. Keep the observed event, possible causes, immediate controls, and eventual corrective action in separate statements so confidence does not outrun evidence.

Briefing checks

Circle one answer in each item. Use the briefing.

1. What was directly observed?

- | | |
|--|---|
| A Connector damage was confirmed by inspection. | B Every sample failed before an hour. |
| C The test setup passed all checks. | D S7 stopped producing an output at minute 46. |

2. Which statement accurately describes the cause?

- | | |
|---|---|
| A The component population is known to be defective. | B Connector damage is an unconfirmed hypothesis. |
| C Connector damage is the established root cause. | D The test fixture has been ruled out. |

3. What cannot be calculated from the supplied facts alone?

- | | |
|---|---|
| A The planned run duration | B Whether the sample reached the full planned duration |
| C A representative population failure rate | D The time of the observed interruption |

Answers and explanations: [page 89](#).

The vocabulary that does the work

B Find the words / 1 of 2

Read the definitions and say the common word combinations aloud.

failure mode

The way an item fails to perform its required function.

IN USE describe the failure mode

failure effect

The consequence of a failure at a stated level.

IN USE assess the failure effect

failure mechanism

The physical or other process producing a failure.

IN USE investigate the failure mechanism

root cause

An underlying cause supported by an appropriate investigation.

IN USE establish the root cause

hypothesis

A proposed explanation that requires testing.

IN USE test a hypothesis

anomaly

An observed departure from expected behavior.

IN USE record an anomaly

intermittent fault

A fault that appears and disappears under some conditions.

IN USE investigate an intermittent fault

reproducibility

Measurement precision under specified changed conditions, such as different laboratories or operators.

IN USE assess reproducibility

repeatability

Agreement of repeated measurements under the same stated conditions.

IN USE check repeatability

test setup

The equipment and arrangement used for a test.

IN USE verify the test setup

test fixture

Equipment that holds or connects the item during testing.

IN USE inspect the test fixture

failure analysis

Investigation of evidence about a failure and its causes.

IN USE commission failure analysis

Precision check

The observed mode is loss of output, not connector damage. A connector problem might explain the event, but it remains a hypothesis. Checking the setup is necessary to interpret the result; it does not justify deleting an inconvenient observation.

The vocabulary that does the work

B Find the words / 2 of 2

Read the definitions and say the common word combinations aloud.

fault isolation

Narrowing a fault to a component, location, or cause.

IN USE support fault isolation

diagnostic evidence

Information used to distinguish possible explanations.

IN USE preserve diagnostic evidence

event log

A time-ordered record of system or test events.

IN USE export the event log

specimen

An identified item submitted for examination or testing.

IN USE preserve the specimen

containment

An immediate measure to limit exposure while investigation continues.

IN USE define containment

corrective action

An action addressing an established cause to prevent recurrence.

IN USE evaluate corrective action

recurrence

Reappearance of a previously observed problem.

IN USE monitor recurrence

FMEA

Failure mode and effects analysis, a structured review of possible failures and consequences.

IN USE update the FMEA

severity

The seriousness of a stated consequence.

IN USE assess severity

occurrence

The likelihood or frequency of a failure cause or event on a defined basis.

IN USE estimate occurrence

detectability

The ability of controls to identify a condition before its consequence.

IN USE assess detectability

residual uncertainty

What remains unknown after available evidence is considered.

IN USE state residual uncertainty

Precision check

Failure mode, effect, and cause answer different questions: what failed, what happened as a consequence, and why. FMEA ratings depend on the chosen method and evidence. Do not invent a universal score or infer a population failure rate from this case.

Phrases you can actually use

C Notice the language / 1 of 2

03

STATE THE OBSERVATION

"S7 stopped producing an output at minute 46."

SEPARATE CAUSE FROM SYMPTOM

"Loss of output is the observed mode, not the confirmed cause."

NAME MISSING CHECKS

"The test setup has not yet been checked."

PROTECT THE RECORD

"Preserve the sample identification and event log."

PRESERVE THE DURATION

"The planned run was 60 minutes."

QUALIFY THE EXPLANATION

"Connector damage is one hypothesis under investigation."

AVOID PREMATURE EXCLUSION

"We have not ruled out a fixture-related explanation."

REQUEST DISCRIMINATING EVIDENCE

"Which observation would distinguish the competing explanations?"

Failed

Needs a required function, condition, and identified item.

Because

Signals a causal claim and needs supporting evidence.

Suspected

Marks an explanation as provisional.

Phrases you can actually use

C Notice the language / 2 of 2

03

KEEP THE RESULT

"An unresolved cause does not erase the observed interruption."

SEPARATE ACTION TYPES

"Containment can precede a confirmed corrective action."

CLARIFY RECURRENCE

"Has the same mode reappeared under documented conditions?"

STATE THE UNCERTAINTY

"The mechanism remains unresolved."

LIMIT THE INFERENCE

"This case does not establish a population failure rate."

AVOID A FALSE FIX

"A replacement alone would not prove the cause was removed."

USE FMEA CAREFULLY

"Link the mode to its possible effects and causes."

CLOSE THE UPDATE

"Report the event, current hypotheses, next checks, and review point."

Could not reproduce

Does not establish that the original event was imaginary.

Contained

Does not mean the underlying cause has been removed.

Fixed

Should be supported by cause-based action and appropriate verification.

Choose the precise message

D Check your understanding

03

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

1. Which draft sentence should replace "connector damage caused the failure"?
 - A The failure rate is one per 46 minutes for every unit.
 - B S7 lost output at minute 46; connector damage remains a hypothesis.
 - C S7 probably worked normally because the setup is unchecked.
 - D All connectors in the product are defective.

2. Why preserve the event log and sample identity?
 - A They establish that every later sample will behave identically.
 - B They connect the investigation to the actual event and configuration.
 - C They automatically prove which component caused the interruption.
 - D They replace all further diagnostic work.

3. A later repeat run shows no interruption. What should happen to the original S7 result?
 - A Confirm connector damage because the event is intermittent.
 - B Declare the test setup fault-free because the repeat ran normally.
 - C Retain the minute-46 event and investigate why the observations differ; one repeat does not identify the cause.
 - D Delete it because the latest run replaces every earlier observation.

4. Which wording distinguishes containment from corrective action?
 - A Every temporary control removes the underlying mechanism.
 - B Exposure controls may begin while the cause-based action remains undecided.
 - C Containment proves that recurrence is impossible.
 - D A suspected cause is enough to declare permanent correction.

Answers and explanations: [page 89](#).

One interruption, several possible explanations

E Conversations / turns 1-10

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

WORD + PHRASE BANK

containment / failure mode / residual uncertainty / corrective action / FMEA / event log / test setup / failure effect / hypothesis / failure analysis

- 01 **Grace:** S7 stopped producing an output at minute forty-six of the planned hour. The draft report says connector damage caused the failure, but we have not examined that explanation yet.
- 02 **Pavel:** Start with the ¹ _____. Loss of output is what we observed. Connector damage is a proposed explanation, and the wording needs to preserve that distinction.
- 03 **Grace:** The team has seen similar symptoms before, which is why the explanation sounds persuasive. I do not want to ignore their experience, but this event has not been reproduced.
- 04 **Pavel:** Treat it as a ² _____. Prior experience can guide the investigation without allowing us to write the conclusion before examining this sample and its records.
- 05 **Grace:** The setup checks are still open too. Can we examine the fixture and connection without treating either as the explanation before we have the evidence?
- 06 **Pavel:** Review the ³ _____ under the approved process. That is part of interpreting the event, not a reason to dismiss the sample interruption as inconvenient.
- 07 **Grace:** I have S7's identifier, the event time, and the configuration details. What other original records should I attach before the team starts the investigation?
- 08 **Pavel:** Preserve the ⁴ _____ as well. We need the original sequence, including anything recorded before the output stopped, rather than a summary written from memory.
- 09 **Grace:** The manager asked whether one failure means the whole population is unreliable. I can report this event, but we do not have the information for that broader conclusion.
- 10 **Pavel:** State the ⁵ _____ plainly. This record does not establish a representative population failure rate, and neither a reassuring nor an alarming estimate should be invented.

The conversation, continued

E Conversations / turns 11-20

03

Continue with the word bank on page 28.

- 11 **Grace:** We should also be clear about what happened beyond the sample. Loss of output is the mode, while the consequence depends on the function it serves.
- 12 **Pavel:** Correct. Describe the ⁶ _____ at the relevant level. Do not let the consequence, the observed mode, and the suspected cause become interchangeable labels.
- 13 **Grace:** Would it be useful to record the possible mode and causes in the structured review now, even while the evidence remains incomplete?
- 14 **Pavel:** Yes, update the ⁷ _____ where appropriate, with the uncertainty visible. Its entries should support investigation and control decisions, not give an unsupported explanation the appearance of formal proof.
- 15 **Grace:** We may need an immediate restriction while the investigation continues. I want to avoid calling that restriction a permanent fix in the status update.
- 16 **Pavel:** Call it ⁸ _____ if that is its actual purpose. The responsible team must define the measure and its limits; the name does not establish that recurrence is prevented.
- 17 **Grace:** Once evidence supports a cause, we can assess a response against it. Simply replacing something and observing a short successful run would not settle every question.
- 18 **Pavel:** A ⁹ _____ needs a cause-based rationale and appropriate verification. Keep the action proposal separate from evidence that it has removed the problem.
- 19 **Grace:** I will revise the report to state the interruption, the unconfirmed connector explanation, the unchecked setup, and the next investigation review. The original result will stay in the record.
- 20 **Pavel:** That supports ¹⁰ _____. We can communicate progress honestly without choosing a cause early or allowing uncertainty to make the observed event disappear.

ANSWERS / EXPLANATIONS: PAGE 90

1. failure mode | 2. hypothesis | 3. test setup | 4. event log | 5. residual uncertainty | 6. failure effect | 7. FMEA | 8. containment | 9. corrective action | 10. failure analysis

Observed does not mean explained

F Say it

Sample K4 loses its signal during a run. A damaged cable is suspected, but neither the cable nor the fixture has been checked. The original log is available.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

hypothesis / mode / log / unresolved

Tester: "Loss of signal is the observed failure ¹ _____."

Engineer: "Cable damage is still a ² _____."

Tester: "We will preserve the original ³ _____."

Engineer: "The root cause remains ⁴ _____."

ANSWERS / EXPLANATIONS: PAGE 90

1. mode | 2. hypothesis | 3. log | 4. unresolved

Rehearse with a fixed script

1. Read the failure discussion. State S7, minute 46, and the planned 60-minute run before naming the connector hypothesis.
2. Switch roles. Separate the observed mode, possible cause, setup checks, containment, and cause-based corrective action.
3. Complete and check the transfer. Preserve K4's original log and keep cable damage unconfirmed.

WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

Three passes do not mean every condition

A Read the situation

Report prototype results with their sample, conditions, criterion, and limits while distinguishing verification from broader validation.

Test lead Ben reviews a presentation with design engineer Amina. Three revision-B prototypes completed a two-hour bench test at 20 degrees Celsius. Their recorded maximum response times were 150, 160, and 175 milliseconds; the test criterion was no more than 200 milliseconds. All three met that criterion. The intended operating range extends from 10 to 40 degrees Celsius, but those other temperatures were not tested. The presentation currently claims that the design is proven for every operating condition.

PEOPLE / Ben: Test lead | Amina: Design engineer

LEAD WITH THE RESULT AND ITS BOUNDARY

A qualified result can still be useful and positive. State what passed before explaining what was not tested. This avoids both overclaiming and burying genuine progress under vague caution. List the exact evidence gap that limits the wider claim.

Briefing checks

Circle one answer in each item. Use the briefing.

1. Which result is supported?

A Every future unit will meet the criterion.

B The design has a proven lifetime reliability of 100%.

C All three prototypes met the stated bench criterion at 20 degrees Celsius.

D The complete operating range was tested.

2. Which value was the largest recorded maximum?

A 160 milliseconds

B 150 milliseconds

C 175 milliseconds

D 200 milliseconds

3. What remains untested in this case?

A The stated two-hour duration at 20 degrees Celsius

B The recorded response-time criterion at 20 degrees Celsius

C Performance at the other temperatures in the intended range

D All three revision-B prototypes

Answers and explanations: page 91.

The vocabulary that does the work

B Find the words / 1 of 2

Read the definitions and say the common word combinations aloud.

test protocol

The documented plan and procedure for a test.

IN USE approve the test protocol

test condition

A specified circumstance under which a test is conducted.

IN USE record test conditions

test article

The identified item subjected to testing.

IN USE identify the test article

sample size

The number of items or observations in a stated sample.

IN USE report the sample size

acceptance limit

The boundary used to judge a measured result.

IN USE compare with the acceptance limit

response time

The time between a defined input and response.

IN USE measure response time

observed maximum

The largest value recorded in the stated observations.

IN USE report the observed maximum

measurement uncertainty

A quantified expression of doubt about a measured value.

IN USE evaluate measurement uncertainty

instrument calibration

Establishing the relation between instrument indications and reference values.

IN USE check instrument calibration

resolution

The smallest distinguishable change in a measurement or indication.

IN USE specify instrument resolution

test coverage

The requirements or conditions addressed by the performed tests.

IN USE identify gaps in test coverage

boundary condition

A limiting input or operating condition used in analysis or testing.

IN USE examine boundary conditions

Precision check

Three out of three is a 100% observed pass fraction for this test. It is not proof of perfect reliability in the population or across the operating range. The largest recorded maximum, 175 ms, is 25 ms below the stated 200 ms criterion.

The vocabulary that does the work

B Find the words / 2 of 2

Read the definitions and say the common word combinations aloud.

verification

Checking that specified requirements have been met.

IN USE complete requirement verification

validation

Assessing whether a product meets intended-use needs.

IN USE plan intended-use validation

representativeness

How well a sample or condition reflects the target use or population.

IN USE assess representativeness

extrapolation

Extending a conclusion beyond the observed range.

IN USE avoid unsupported extrapolation

confidence interval

A statistical interval constructed by a method with stated coverage properties.

IN USE report a confidence interval

pass fraction

The proportion of tested items meeting a stated criterion.

IN USE calculate the observed pass fraction

population

The full set of items about which an inference is intended.

IN USE define the target population

censored observation

A lifetime observation ending before the event of interest occurs.

IN USE account for censored observations

test duration

The length of time a test runs.

IN USE state the test duration

reliability claim

A statement about performance without failure under stated conditions over time.

IN USE qualify a reliability claim

test deviation

A recorded departure from the approved test procedure.

IN USE assess a test deviation

evidence boundary

The limit of what the available evidence establishes.

IN USE state the evidence boundary

Precision check

Verification checks specified requirements; validation addresses intended-use needs. Passing one bench criterion does not settle both. No uncertainty budget, statistical sampling plan, or lifetime model is supplied here, so do not invent a confidence level or a life prediction.

Phrases you can actually use

C Notice the language / 1 of 2

LEAD WITH THE RESULT

"All three prototypes met the stated criterion."

GIVE THE CONDITIONS

"The test ran for two hours at 20 degrees Celsius."

REPORT THE LARGEST VALUE

"The largest recorded maximum was 175 milliseconds."

BOUND THE PERCENTAGE

"The observed pass fraction was three out of three."

NAME THE CONFIGURATION

"These results apply to revision B."

STATE THE CRITERION

"The response-time limit was 200 milliseconds."

PRESERVE THE UNIT

"These values are in milliseconds, not seconds."

AVOID POPULATION OVERCLAIM

"That is not proof that every production unit will pass."

Passed

Always attach the criterion and tested configuration.

Proven

Usually overbroad without a clearly bounded claim.

100%

Can describe a sample proportion without proving population certainty.

Phrases you can actually use

C Notice the language / 2 of 2

IDENTIFY MISSING COVERAGE

"The other intended temperatures were not tested."

QUALIFY THE HEADLINE

"Replace proven everywhere with the result actually demonstrated."

KEEP RECORDS CONNECTED

"Link the result to its test article and protocol."

PRESERVE POSITIVE PROGRESS

"The test supports this criterion under these conditions."

SEPARATE PURPOSES

"This bench check is not complete intended-use validation."

ASK ABOUT UNCERTAINTY

"Where is the measurement-uncertainty assessment?"

AVOID INVENTED CONFIDENCE

"The supplied data do not provide a stated confidence level."

CLOSE THE REPORT

"State the result, conditions, limitation, and next evidence need."

Maximum

Specify whether observed, allowed, or predicted.

Representative

Requires a reason why the sample reflects intended use.

Validation

Do not use it as a synonym for any successful bench test.

Choose the precise message

D Check your understanding

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

1. Which presentation headline is most accurate?

- A** Revision B is proven across all operating conditions.
- B** Every manufactured unit will respond within 175 milliseconds.
- C** Two hours establish the complete service life.
- D** Three revision-B prototypes met the response-time criterion in the stated bench test.

2. What does the observed 100% pass fraction mean?

- A** All three tested items met this criterion in this test.
- B** The true population failure probability is known to be zero.
- C** The entire 10-to-40-degree range has been covered.
- D** No additional evidence could change the assessment.

3. The largest recorded maximum is 175 ms against a 200 ms limit. Which statement is precise?

- A** The observed gap to the limit is 25 ms; no supplied uncertainty budget establishes a guaranteed margin for all units.
- B** The product is guaranteed to remain 25 ms below the limit at every temperature.
- C** The limit should be revised to 175 ms because that is the largest observed value.
- D** The gap is 25 seconds, proving the five-year service target.

4. Which limitation is specific and useful?

- A** The design necessarily fails at every untested temperature.
- B** All intended-use requirements are automatically satisfied.
- C** Performance at the other intended temperatures was not tested.
- D** Nothing at all can be learned from the test.

Answers and explanations: [page 91](#).

Three passes do not mean every condition

E Conversations / turns 1-10

04

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

WORD + PHRASE BANK

evidence boundary / extrapolation / sample size / reliability claim / observed maximum / test conditions / pass fraction / verification / validation / test coverage

- 01 **Amina:** The presentation says the design is proven for every operating condition. All three prototypes passed, so I thought that was a concise way to report the positive result.
- 02 **Ben:** We can report the ¹ _____, but the headline goes further. Three out of three describes the tested sample, not every unit or every condition in the operating range.
- 03 **Amina:** Let us put the actual conditions beside the result. These were revision-B prototypes, tested for two hours at twenty degrees Celsius under the bench procedure.
- 04 **Ben:** Those ² _____ are essential. Readers should not have to search an appendix to discover that the rest of the intended temperature range was not tested.
- 05 **Amina:** The recorded maximum response times were one hundred fifty, one hundred sixty, and one hundred seventy-five milliseconds. The criterion allowed up to two hundred milliseconds.
- 06 **Ben:** Then the largest ³ _____ was one hundred seventy-five. That is twenty-five milliseconds below the criterion, but it is not a universal product limit.
- 07 **Amina:** I will write three out of three met the criterion. That is clearer than a large one-hundred-percent headline with the sample size hidden below it.
- 08 **Ben:** Exactly. Keep the ⁴ _____ visible. A percentage can sound much more comprehensive than the underlying evidence when the number of tested items is omitted.
- 09 **Amina:** The intended operating range is ten to forty degrees. Since we tested only twenty, the other temperatures remain a gap rather than a demonstrated pass.
- 10 **Ben:** That is a ⁵ _____ limitation. State it directly; do not imply the untested temperatures failed, but do not label them verified either.

The conversation, continued

E Conversations / turns 11-20

04

Continue with the word bank on page 37.

- 11 **Amina:** Would it help to distinguish checking this performance criterion from confirming that the product works adequately for its intended users and operating tasks?
- 12 **Ben:** Yes. This supports ⁶ _____ of the stated criterion under the recorded conditions. Broader intended-use questions need their own evidence and cannot be settled by relabeling this test.
- 13 **Amina:** The slide also calls the product fully validated. We have not supplied evidence for the complete use context, so that phrase creates a second overclaim.
- 14 **Ben:** Remove the unsupported ⁷ _____ claim. We can explain what has been demonstrated and which intended-use evidence remains outstanding without making the result sound like a failure.
- 15 **Amina:** What should I say if someone asks about ten and forty degrees? We have no supporting model or test at those temperatures in this package.
- 16 **Ben:** Then that would be unsupported ⁸ _____. Do not substitute a plausible expectation for evidence, especially when the presentation may guide a release or customer commitment.
- 17 **Amina:** I will also avoid adding a confidence percentage. The supplied records do not include a statistical sampling plan, uncertainty budget, or lifetime analysis.
- 18 **Ben:** Correct. A ⁹ _____ needs its own defined basis. The fact that no failure occurred in these runs does not establish a zero population failure probability.
- 19 **Amina:** The revised headline will say three revision-B prototypes met the response-time criterion during a two-hour bench test at twenty degrees. The next sentence will identify the untested temperatures.
- 20 **Ben:** That states the ¹⁰ _____ clearly. We retain the positive result and give the review team enough context to decide what additional evidence is needed.

ANSWERS / EXPLANATIONS: PAGE 92

1. pass fraction | 2. test conditions | 3. observed maximum | 4. sample size | 5. test coverage | 6. verification | 7. validation | 8. extrapolation | 9. reliability claim | 10. evidence boundary

Four passes, one tested temperature

F Say it

Four prototypes pass a three-hour bench test at 25 degrees Celsius. The intended range also includes lower and higher temperatures, which were not tested. No population reliability estimate is supplied.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

untested / reliability / hours / four

Engineer: "The tested sample size was ¹ _____."

Reviewer: "The duration was three ² _____."

Engineer: "The other temperatures remain ³ _____."

Reviewer: "The result does not establish population ⁴ _____."

ANSWERS / EXPLANATIONS: PAGE 92

1. four | 2. hours | 3. untested | 4. reliability

Rehearse with a fixed script

1. Read the test-results dialogue. Say three prototypes, revision B, two hours, 20 degrees Celsius, and the 200 ms criterion.
2. Switch roles. Bound the 175 ms observed maximum and 100% sample pass fraction without extending them across 10-40 degrees Celsius.
3. Complete and check the transfer. Keep four passes at 25 degrees Celsius separate from untested temperatures and population reliability.

WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

Cheaper material, more expensive assembly

A Read the situation

Compare design costs on the same volume and cost basis while keeping manufacturability and approval separate from arithmetic.

Design engineer Tessa and manufacturing engineer Arun compare two proposed housings for 3,000 acceptable units. Option A has material cost of \$8 and assembly cost of \$6 per unit, plus \$1,000 one-time tooling. Option B has material cost of \$10 and assembly cost of \$2 per unit, plus \$5,000 tooling. These are the only cost elements in the comparison, with no scrap assumed. Technical suitability, supplier capacity, and approval are unresolved. A slide nevertheless recommends A solely because its material is cheaper.

PEOPLE / Tessa: Design engineer | Arun: Manufacturing engineer

COMPARE LIKE WITH LIKE

An engineer can challenge a cost recommendation without challenging the cost-saving objective. Put both alternatives on the same quantity, cost categories, and assumptions. Then distinguish the numerical result from the separate evidence needed to select and release a design.

Briefing checks

Circle one answer in each item. Use the briefing.

1. What is the recurring material-plus-assembly cost of A?

- | | |
|------------------------|------------------------|
| A \$8 per unit | B \$6 per unit |
| C \$12 per unit | D \$14 per unit |

2. What is the total stated cost of B for 3,000 units?

- | | |
|-------------------|-------------------|
| A \$36,000 | B \$35,000 |
| C \$47,000 | D \$41,000 |

3. What remains unresolved after the calculation?

- | | |
|--|--|
| A The supplied tooling amount for A | B The supplied material cost for B |
| C The stated comparison quantity | D Technical suitability, capacity, and approval |

Answers and explanations: [page 93](#).

The vocabulary that does the work

B Find the words / 1 of 2

Read the definitions and say the common word combinations aloud.

manufacturability

How readily a design can be produced by the intended process.

IN USE assess manufacturability

design for assembly

Design choices that simplify or improve assembly, abbreviated DFA.

IN USE apply design for assembly

design for manufacture

Design choices suited to the intended manufacturing process, abbreviated DFM.

IN USE review design for manufacture

bill of materials

The structured list of parts and materials in a product, abbreviated BOM.

IN USE compare the bill of materials

material cost

The stated cost of materials used in a unit.

IN USE estimate material cost

assembly cost

The cost assigned to assembling a unit.

IN USE reduce assembly cost

recurring cost

A cost incurred repeatedly as units are produced.

IN USE compare recurring cost

nonrecurring cost

A cost incurred once for the defined program or setup.

IN USE identify nonrecurring cost

tooling investment

Expenditure on equipment or tools needed for production.

IN USE justify tooling investment

unit cost

Cost per unit on a stated inclusion and volume basis.

IN USE calculate unit cost

cost model

A structured calculation connecting cost assumptions and totals.

IN USE review the cost model

cost driver

A factor that materially changes cost.

IN USE identify the cost driver

Precision check

A costs \$43,000 and B costs \$41,000 for 3,000 units on the stated basis. B saves \$2 per unit in recurring cost but requires \$4,000 more tooling. Their totals are equal at 2,000 units. These arithmetic results do not establish technical suitability.

The vocabulary that does the work

B Find the words / 2 of 2

Read the definitions and say the common word combinations aloud.

production volume

The number of units made in a stated period or program.

IN USE state production volume

break-even volume

The quantity at which compared total costs are equal.

IN USE calculate break-even volume

incremental cost

The change in cost between specified alternatives.

IN USE compare incremental cost

cost trade-off

An exchange between cost categories or other design outcomes.

IN USE explain the cost trade-off

process capability

The ability of a stable process to meet specified limits.

IN USE assess process capability

cycle time

Time taken for one defined production cycle.

IN USE measure cycle time

takt time

Available production time divided by required customer demand.

IN USE compare cycle time with takt time

scrap allowance

A stated cost or quantity provision for unusable production.

IN USE include a scrap allowance

yield assumption

The assumed proportion of production meeting a defined criterion.

IN USE challenge the yield assumption

tolerance relaxation

Widening a dimensional limit subject to design requirements.

IN USE assess tolerance relaxation

part consolidation

Combining functions to reduce the number of separate parts.

IN USE evaluate part consolidation

make-or-buy decision

Choosing between internal production and external supply.

IN USE support a make-or-buy decision

Precision check

Unit cost can include different categories. Name the categories before comparing figures. This case excludes other cost elements and assumes no scrap; those simplifying assumptions must not be presented as a complete real-world business case.

Phrases you can actually use

C Notice the language / 1 of 2

SET THE COMPARISON

"Use the same 3,000-unit quantity for both options."

CHALLENGE THE NARROW CLAIM

"Lower material cost does not necessarily mean lower total cost."

STATE THE ALTERNATIVE

"B has \$12 of recurring cost per unit."

GIVE THE TOTAL

"The stated totals are \$43,000 for A and \$41,000 for B."

NAME THE CATEGORIES

"Include material, assembly, and one-time tooling."

ADD RECURRING COSTS

"A has \$14 of material and assembly cost per unit."

KEEP TOOLING SEPARATE

"B requires \$4,000 more upfront tooling."

EXPLAIN THE DIFFERENCE

"B is \$2,000 lower at this volume."

Cheaper

Needs a cost boundary and quantity.

Per unit

Can mean recurring cost or a total allocated over volume.

Upfront

Describes timing, not automatically recoverability.

Phrases you can actually use

C Notice the language / 2 of 2

IDENTIFY THE CROSSOVER

"The total costs are equal at 2,000 units."

EXPOSE THE ASSUMPTION

"No scrap is assumed in these figures."

ASK ABOUT CAPACITY

"Can the proposed process deliver the required volume?"

CHECK THE DENOMINATOR

"Which production quantity supports that unit-cost figure?"

QUALIFY THE MODEL

"The comparison includes only the listed cost elements."

PROTECT TECHNICAL CRITERIA

"A cost advantage does not establish manufacturability."

AVOID PREMATURE SELECTION

"B leads on this cost basis, pending the other reviews."

CLOSE THE RECOMMENDATION

"Separate the cost result from the design approval decision."

Break-even

Here means equal option totals, not business profitability.

No scrap

A model assumption, not proof of perfect production yield.

Preferred

State the criterion on which the preference is based.

Choose the precise message

D Check your understanding

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

1. Which recommendation is supported at 3,000 units?

- A** A and B have equal totals at every quantity.
- B** B has the lower stated cost, subject to unresolved technical and capacity reviews.
- C** A must be chosen because its material costs less.
- D** B is technically approved because its calculated total is lower.

2. At which volume are the stated totals equal?

- A** 3,000 units
- B** 4,000 units
- C** 2,000 units
- D** 500 units

3. B saves \$2 per unit for 3,000 units. Why is the stated total saving \$2,000 rather than \$6,000?

- A** The \$6,000 recurring saving is offset by B's \$4,000 additional tooling.
- B** B's \$5,000 tooling should be deducted twice.
- C** A's tooling is recurring and must be multiplied by 3,000.
- D** The two-dollar difference applies only to the first 1,000 units.

4. Why should the model state no scrap?

- A** Scrap can never affect unit cost.
- B** Different production yields could change the cost comparison.
- C** The assumption proves every real unit will be acceptable.
- D** Tooling cost disappears when scrap is excluded.

Answers and explanations: [page 93](#).

Cheaper material, more expensive assembly

E Conversations / turns 1-10

05

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

WORD + PHRASE BANK

process capability / break-even volume / cost model / unit cost / yield assumption / cost trade-off / production volume / tooling investment / recurring cost / manufacturability

- 01 **Tessa:** The slide recommends A because its material costs eight dollars per unit instead of ten. I have not included assembly or tooling in that recommendation yet.
- 02 **Arun:** Then we need the full stated ¹ _____ before recommending an option. Use material, assembly, and one-time tooling for three thousand acceptable units, with no scrap.
- 03 **Tessa:** A is eight plus six, so fourteen per unit. B is ten plus two, so twelve. The cheaper material does not give A the cheaper repeated work.
- 04 **Arun:** Correct. B saves two dollars in ² _____ per unit. That comparison includes assembly but not either option's one-time production equipment expense.
- 05 **Tessa:** Two dollars times three thousand gives six thousand saved. Should I put six thousand as the overall saving on the slide?
- 06 **Arun:** Not yet. B's ³ _____ is five thousand against A's one thousand. Subtract that additional four thousand from the recurring saving; the overall difference is two thousand.
- 07 **Tessa:** Let me check the totals: A is forty-two thousand plus one thousand, and B is thirty-six thousand plus five thousand. Forty-three versus forty-one thousand.
- 08 **Arun:** Yes, at the stated ⁴ _____ of three thousand. Attach the quantity to the conclusion; a smaller order may not recover B's extra upfront expense.
- 09 **Tessa:** The extra tooling is four thousand and the recurring difference is two per unit. Dividing gives two thousand units. Is that where the costs are equal?
- 10 **Arun:** That is the ⁵ _____. At two thousand units, A is twenty-eight thousand plus one thousand; B is twenty-four thousand plus five thousand. Both total twenty-nine thousand.

The conversation, continued

E Conversations / turns 11-20

Continue with the word bank on page 46.

- 11 **Tessa:** At only one thousand units, A becomes fifteen thousand and B seventeen thousand. So saying B is always cheaper would also be wrong.
- 12 **Arun:** Exactly. State the ⁶ _____: higher tooling buys lower recurring expense. The comparison changes with quantity even before we examine any unlisted costs.
- 13 **Tessa:** Can we recommend buying B now that it wins this calculation? Its assembly estimate is attractive, but technical suitability has not been reviewed.
- 14 **Arun:** No. Its ⁷ _____ is still unresolved. The cost sheet does not show that the proposed process can produce a housing meeting the design requirements.
- 15 **Tessa:** I also need the supplier's capacity confirmation. Having enough machines available would answer volume, but not necessarily whether the parts meet dimensional limits.
- 16 **Arun:** Right. Assess ⁸ _____ separately from capacity. Consistent conformance under a stable process and available production quantity are different questions.
- 17 **Tessa:** We assumed every produced unit is acceptable. I will state that clearly, rather than present the model as though we measured the supplier's losses.
- 18 **Arun:** Keep that ⁹ _____ beside the totals. Scrap or rework could change acceptable-unit cost; the present comparison includes only the three listed cost categories.
- 19 **Tessa:** I will revise the slide: B is two thousand lower at three thousand units on the stated basis, pending suitability, capacity, and approval. No purchase is authorized.
- 20 **Arun:** Also label each ¹⁰ _____ figure as recurring-only or tooling-inclusive. That will stop the next reader comparing twelve dollars with a total-cost figure from another volume.

ANSWERS / EXPLANATIONS: PAGE 94

1. cost model | 2. recurring cost | 3. tooling investment | 4. production volume | 5. break-even volume | 6. cost trade-off | 7. manufacturability | 8. process capability | 9. yield assumption | 10. unit cost

When the quantity changes the answer

F Say it

Option C costs \$9 per unit plus \$2,000 tooling. Option D costs \$7 per unit plus \$6,000 tooling. Only these costs apply, with no scrap. Compare 1,000 units.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

2000 / C / 13000 / 11000

Engineer: "C totals ¹ _____ dollars."

Reviewer: "D totals ² _____ dollars."

Engineer: "At this quantity, the lower-cost option is ³ _____."

Reviewer: "Equal totals occur at ⁴ _____ units."

ANSWERS / EXPLANATIONS: PAGE 94

1. 11000 | 2. 13000 | 3. C | 4. 2000

Rehearse with a fixed script

1. Read turns 1-10. Correct the \$6,000 recurring saving to \$2,000 overall after the extra tooling, then check the 2,000-unit crossover.
2. Switch roles for turns 11-20. Separate cost, capacity, process capability, and design approval.
3. Complete and check the transfer. Calculate C at \$11,000 and D at \$13,000 for 1,000 units, then state the 2,000-unit crossover.

WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

A ratio is not a compliance decision

A Read the situation

Question a compliance claim by identifying the applicable requirement, analysis basis, evidence scope, and decision authority.

Engineer Rosa and assurance lead Malik review a fictional design note labeled compliant. A preliminary static model reports a safety factor of 2.0 using its stated capacity-to-demand convention. The applicable requirement, acceptance threshold, and reviewer are not identified. Fatigue and dynamic loading are outside the model. No release approval or formal conformity assessment is recorded. Rosa must correct the note without claiming the design is either safe or unsafe from the ratio alone.

PEOPLE / Rosa: Design engineer | Malik: Assurance lead

06

ASK WHAT COMPLIANT REFERS TO

Compliance is a relationship to specific applicable requirements, not a general compliment about design quality. Ask for the requirement, version, evidence, limitations, and authorized decision. A numerical margin may be useful evidence, but its meaning depends on the model and the failure modes considered.

Briefing checks

Circle one answer in each item. Use the briefing.

1. What does 2.0 describe in the note?

A A preliminary static-model ratio under its stated convention

C A demonstrated fatigue life

B Complete product conformity

D An authorized release decision

2. Which behavior is outside the model's assessment?

A The model's stated capacity-to-demand convention

C Fatigue and dynamic-loading effects

B The numerical result of 2.0

D Every possible static input

3. What is missing from the compliance claim?

A The applicable requirement and acceptance threshold

C A statement that the model is static

B A preliminary ratio of 2.0

D A statement that release approval is absent

Answers and explanations: [page 95](#).

The vocabulary that does the work

B Find the words / 1 of 2

Read the definitions and say the common word combinations aloud.

factor of safety

A defined ratio comparing capacity and demand on a stated basis.

IN USE state the factor of safety

allowable

A permitted value established by the applicable design basis.

IN USE identify the allowable

design load

A load case specified for design assessment.

IN USE define the design load

load case

A defined combination of actions considered in an analysis.

IN USE assess the load case

static analysis

Analysis of conditions treated without significant time-dependent dynamics.

IN USE review static analysis

dynamic loading

Loading whose time variation requires dynamic effects to be considered.

IN USE assess dynamic loading

fatigue

Progressive damage associated with repeated loading.

IN USE evaluate fatigue

ultimate capacity

Capacity associated with a defined ultimate failure limit.

IN USE state ultimate capacity

yield limit

A boundary associated with the onset of specified permanent deformation.

IN USE identify the yield limit

buckling

Instability involving a change in structural form under loading.

IN USE assess buckling

failure criterion

The condition used to define failure in an analysis.

IN USE specify the failure criterion

load combination

A defined set of loads considered together.

IN USE document the load combination

Precision check

A factor of safety is meaningful only with its stated convention, failure criterion, inputs, and load cases. A value of 2.0 in one preliminary static model is not a universal safety threshold and cannot establish fatigue performance or complete conformity.

The vocabulary that does the work

B Find the words / 2 of 2

Read the definitions and say the common word combinations aloud.

model validity

The degree to which a model is suitable for its intended assessment.

IN USE assess model validity

analysis limitation

A condition or behavior not adequately addressed by an analysis.

IN USE state analysis limitations

applicable requirement

A requirement that governs the specific product or situation.

IN USE identify applicable requirements

compliance matrix

A record linking requirements to evidence and status.

IN USE maintain a compliance matrix

conformity assessment

Evaluation of whether specified requirements are fulfilled.

IN USE identify the conformity assessment route

normative reference

A referenced document needed to apply a specification's requirements.

IN USE check normative references

standard edition

The identified version of a published standard.

IN USE confirm the standard edition

evidence package

The records supporting a technical or compliance conclusion.

IN USE assemble the evidence package

independent review

Review by a suitably separate qualified reviewer.

IN USE arrange independent review

authorized sign-off

Recorded approval by a person with the relevant authority.

IN USE obtain authorized sign-off

deviation approval

Authorized acceptance of a defined departure where permitted.

IN USE record deviation approval

release restriction

A stated limit on what may proceed under a decision.

IN USE communicate release restrictions

Precision check

Requirements may come from law, contracts, standards, and internal specifications, with different authority and scope. Identify what actually applies. Neither a missing compliance record nor a favorable ratio alone justifies declaring the product categorically safe or unsafe.

Phrases you can actually use

C Notice the language / 1 of 2

ASK FOR THE BASIS

"Compliant with which applicable requirement?"

NAME THE CONVENTION

"The reported factor uses the stated capacity-to-demand ratio."

ASK FOR THE CRITERION

"Which failure criterion and acceptance threshold apply?"

AVOID A UNIVERSAL RULE

"A factor of 2.0 is not a universal acceptance threshold."

IDENTIFY THE VERSION

"Which standard edition does the assessment use?"

LIMIT THE RESULT

"This is a preliminary static-model result."

EXPOSE EXCLUDED BEHAVIOR

"Fatigue and dynamic loading are outside this model."

SEPARATE ANALYSIS FROM APPROVAL

"A calculated result is not authorized sign-off."

06

Safe

A broad claim that cannot rest on one unexplained ratio.

Compliant

Requires an identified applicable requirement and evidence.

Factor

Specify the ratio convention and relevant failure criterion.

Phrases you can actually use

C Notice the language / 2 of 2

REQUEST TRACEABILITY

"Link each requirement to its supporting evidence."

KEEP STATUS PRECISE

"The compliance conclusion is not yet established."

ASK ABOUT REVIEW

"Who is qualified and authorized to review this assessment?"

PRESERVE RESTRICTIONS

"State what cannot proceed while the item remains open."

CHECK MODEL SCOPE

"Does the model represent the load cases being claimed?"

AVOID UNSUPPORTED ALARM

"The missing record does not itself prove physical failure."

CLARIFY A DEPARTURE

"Any permitted deviation needs the required approval."

CLOSE THE NOTE

"Report the result, limits, missing basis, and pending decision."

06

Preliminary

Signals a maturity limit, not a release authorization.

Outside scope

Makes excluded behavior visible to the reader.

Signed

Verify the signer's authority and the exact decision covered.

Choose the precise message

D Check your understanding

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

1. Which replacement note is justified?

- A** The product is fully compliant because 2.0 exceeds one.
- B** The product is proven unsafe because no reviewer is listed.
- C** Fatigue is acceptable because the static result is favorable.
- D** The preliminary static model reports 2.0; applicable acceptance requirements and review remain unresolved.

2. Why is the applicable requirement necessary?

- A** It establishes what must be assessed and the acceptance basis.
- B** It changes every preliminary result into automatic approval.
- C** It removes the need to identify model limitations.
- D** It guarantees that all standards use the same factor convention.

3. The static factor is 2.0 and fatigue is outside the model. Which conclusion is justified?

- A** Fatigue can be ignored once the static factor exceeds one.
- B** Fatigue performance is not established by this model; it needs an appropriate assessment basis.
- C** Fatigue life is doubled because the factor is two.
- D** Fatigue has failed because the report omits it.

4. What should a compliance matrix connect?

- A** Every possible standard regardless of relevance
- B** A release signature with no stated scope
- C** Applicable requirements, evidence, and assessment status
- D** Only the project team's preferred conclusion

Answers and explanations: page 95.

A ratio is not a compliance decision

E Conversations / turns 1-10

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

WORD + PHRASE BANK

failure criterion / compliance matrix / factor of safety / applicable requirement / standard edition / model validity / conformity assessment / analysis limitations / authorized sign-off / evidence package

06

- 01 **Rosa:** The note says compliant, and the preliminary analysis reports a factor of two. Before I circulate it, I want to check whether that label is justified.
- 02 **Malik:** Start with the ¹ _____. Compliant with what, and on which acceptance basis? A favorable number does not identify the requirement or the scope of the conclusion.
- 03 **Rosa:** The model uses its stated capacity-to-demand convention. It is a static assessment, but the note does not identify the required threshold or who reviewed the result.
- 04 **Malik:** Then describe the ² _____ on that limited basis. Two is the reported preliminary value, not a universal rule that establishes compliance for every design.
- 05 **Rosa:** I will name the failure condition beside the ratio. Which inputs and assumptions must the reviewer see to understand what this particular calculation covers?
- 06 **Malik:** Exactly. Name the ³ _____ and the relevant inputs. A ratio without that context can sound precise while leaving its actual technical meaning unclear.
- 07 **Rosa:** Fatigue and dynamic loading were excluded. I will state that directly instead of allowing the static result to stand in for those separate questions.
- 08 **Malik:** Those are important ⁴ _____. The reader needs them beside the result, especially if the note might be used to support a broader design or release decision.
- 09 **Rosa:** Could I say the static result is encouraging, provided I avoid saying it proves the product safe? I do not want to hide useful work either.
- 10 **Malik:** Report the result, but keep ⁵ _____ and review status explicit. The model must be suitable for the claim, and the acceptance basis still needs to be identified.

The conversation, continued

E Conversations / turns 11-20

Continue with the word bank on page 55.

- 11 **Rosa:** I will check the contract requirements too. Can we identify the applicable documents and versions before the note lists standards that may not govern this product?
- 12 **Malik:** Yes, including the relevant ⁶ _____ where a standard applies. Do not list an impressive collection of references without establishing their relevance and authority.
- 13 **Rosa:** I can organize the record so each applicable requirement points to the analysis or test evidence and shows whether assessment is complete, open, or awaiting review.
- 14 **Malik:** That is the purpose of a ⁷ _____. It should expose the missing links rather than turn a single favorable calculation into a blanket green status.
- 15 **Rosa:** We also have no formal conformity assessment recorded. I should not write as though assembling the internal note has completed whatever external or internal route applies.
- 16 **Malik:** Correct. The ⁸ _____ route needs confirmation for this product and jurisdiction. This discussion cannot replace the responsible specialists or the required process.
- 17 **Rosa:** Then the revised note can preserve the factor of two, identify the model exclusions, and say that the compliance conclusion remains unestablished pending the required review.
- 18 **Malik:** Also distinguish that from ⁹ _____. No release approval is recorded, and a technical result must not silently become permission to proceed.
- 19 **Rosa:** I will keep the current restrictions visible and identify who will confirm the requirements and review the evidence. I will not call the design either safe or unsafe from this ratio alone.
- 20 **Malik:** Good. The ¹⁰ _____ should let a qualified decision maker see what is supported and what is missing. That is more useful than an unsupported compliance label.

06

ANSWERS / EXPLANATIONS: PAGE 96

1. applicable requirement | 2. factor of safety | 3. failure criterion | 4. analysis limitations | 5. model validity | 6. standard edition | 7. compliance matrix | 8. conformity assessment | 9. authorized sign-off | 10. evidence package

Keep the model inside its scope

F Say it

A preliminary static report gives a factor of 1.8 under its stated convention. It excludes fatigue, has no identified acceptance threshold, and has not received release approval.

Complete the four gaps. Use each bank entry once.

06

WORD + PHRASE BANK

granted / fatigue / unidentified / static

Engineer: "The reported value belongs to the ¹ _____ model."

Reviewer: "The model excludes ² _____."

Engineer: "The acceptance threshold remains ³ _____."

Reviewer: "Release approval has not been ⁴ _____."

ANSWERS / EXPLANATIONS: PAGE 96

1. static | 2. fatigue | 3. unidentified | 4. granted

Rehearse with a fixed script

1. Read the assurance discussion. Keep the reported 2.0 factor within its preliminary static-model basis.
2. Switch roles. Name the unidentified requirement and excluded fatigue and dynamic effects without declaring universal safety or failure.
3. Complete and check the transfer. Read the 1.8 factor, missing threshold, and absent release approval as separate facts.

WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

Correct the cause without losing the customer

A Read the situation

Correct an unsupported causal statement, acknowledge customer impact, and give a useful next update without promising an unconfirmed resolution.

A customer reports two output interruptions from field unit L18. Service engineer Marco previously emailed that humidity caused the interruptions, although humidity remains only a hypothesis and no root cause is confirmed. Customer engineer Nia requests a correction and a resolution date. The investigation is open; records are being gathered, no repair or operating change is authorized, and Marco can commit only to an update by 10 a.m. on October 2. The cases are fictional and supply no instruction to continue operating the unit.

PEOPLE / Nia: Customer engineer | Marco: Service engineer

07

CORRECT THE RECORD EXPLICITLY

A quiet change of wording may leave the customer relying on the earlier claim. Identify what was overstated, give the corrected status, and acknowledge the practical effect of uncertainty. A promised update is valuable when it specifies what will be reported, even if resolution timing remains unknown.

Briefing checks

Circle one answer in each item. Use the briefing.

1. What needs correction?

- | | |
|---|--|
| A Humidity was stated as a confirmed cause without confirmation. | B The customer reported only one interruption. |
| C A repair was authorized but not recorded. | D The investigation was completed before the email. |

2. What can Marco commit to?

- | | |
|---|---|
| A An update by 10 a.m. on October 2 | B A completed repair by 10 a.m. on October 2 |
| C A confirmed cause before any record review | D Unrestricted continued operation |

3. Which statement preserves the customer's report?

- | | |
|--|--|
| A Every unit has a humidity defect. | B L18 is known to be unaffected. |
| C The interruptions did not occur because the cause is unknown. | D Unit L18 has two reported output interruptions. |

Answers and explanations: [page 97](#).

The vocabulary that does the work

B Find the words / 1 of 2

Read the definitions and say the common word combinations aloud.

field report

Information about product behavior in actual use.

IN USE assess a field report

customer impact

The effect of an issue on the customer's work or results.

IN USE acknowledge customer impact

reported symptom

Behavior described by a user but not necessarily independently verified.

IN USE record the reported symptom

confirmed finding

A conclusion supported by the completed relevant checks.

IN USE separate confirmed findings

working hypothesis

A provisional explanation guiding investigation.

IN USE qualify the working hypothesis

causal statement

Wording that says one condition produced another.

IN USE correct a causal statement

correction notice

A message explicitly amending earlier information.

IN USE issue a correction notice

case reference

An identifier connecting records for the same issue.

IN USE use the case reference

unit serial number

A unique identifier for an individual product unit.

IN USE confirm the unit serial number

installed configuration

The hardware, software, and settings present in the field.

IN USE document the installed configuration

operating history

Records of how and when a unit has been used.

IN USE request operating history

evidence request

A specific request for information needed in an investigation.

IN USE make an evidence request

Precision check

Reported interruptions are not erased by uncertainty about cause. The error is the email saying humidity caused them. The correction should explicitly withdraw that unsupported conclusion while preserving humidity as a hypothesis and the customer report as relevant evidence.

The vocabulary that does the work

B Find the words / 2 of 2

Read the definitions and say the common word combinations aloud.

traceable record

A record connected to its source, time, and relevant item.

IN USE retain a traceable record

return authorization

Permission and arrangements for returning an item.

IN USE confirm return authorization

chain of custody

A record of possession and transfer of evidence or an item.

IN USE maintain chain of custody

service instruction

An authorized direction for servicing or handling a product.

IN USE follow the service instruction

field action

A coordinated response affecting products in use.

IN USE assess a field action

impact statement

A bounded description of an issue's effects.

IN USE prepare an impact statement

interim update

A progress report before the final outcome is available.

IN USE provide an interim update

resolution commitment

A promise about when or how an issue will be resolved.

IN USE avoid an unsupported resolution commitment

update cadence

The agreed frequency or timing of progress communication.

IN USE agree an update cadence

escalation contact

The named route for raising concerns requiring further attention.

IN USE identify the escalation contact

case owner

The person responsible for coordinating a case.

IN USE name the case owner

closure confirmation

A recorded check that the case's closure conditions are met.

IN USE obtain closure confirmation

Precision check

An update commitment is not a repair deadline. State what the update will contain, even if the investigation remains open. Do not improvise operating or repair instructions; any such direction must come through the authorized process for the actual product and risk.

Phrases you can actually use

C Notice the language / 1 of 2

ACKNOWLEDGE THE CORRECTION

"Our earlier email overstated the cause."

PRESERVE THE HYPOTHESIS

"It remains one explanation under investigation."

ACKNOWLEDGE THE IMPACT

"I understand that the uncertainty affects your planning."

AVOID FALSE REASSURANCE

"We do not yet have evidence to rule out recurrence."

WITHDRAW THE UNSUPPORTED CLAIM

"Humidity has not been confirmed as the cause."

RECOGNIZE THE REPORT

"We have recorded two interruptions on unit L18."

ASK FOR SPECIFIC RECORDS

"Please connect each event to its time and installed configuration."

SEPARATE THE COMMITMENTS

"I can commit to an update, not a resolution date."

Reported

Attributes a statement without dismissing it.

Caused

Requires stronger evidence than a plausible association.

Correction

Explicitly identifies the earlier information that changed.

Phrases you can actually use

C Notice the language / 2 of 2

STATE THE NEXT CONTACT

"I will update you by 10 a.m. on October 2."

KEEP AUTHORITY CLEAR

"No repair or operating change is authorized by this message."

CONNECT THE CORRECTION

"Attach this correction to the same case record."

AVOID PREMATURE CLOSURE

"Sending the correction does not close the investigation."

DEFINE UPDATE CONTENTS

"The update will distinguish findings, open questions, and next steps."

ROUTE SERVICE DECISIONS

"Product-specific instructions must come through the authorized service process."

NAME OWNERSHIP

"I remain the case owner for communication."

CLOSE THE LOOP

"Please confirm that the corrected status reached the relevant team."

07

Update

A communication milestone, not necessarily a technical resolution.

No evidence yet

Does not mean proof of absence.

Closed

Needs the case's actual closure conditions to be satisfied.

Choose the precise message

D Check your understanding

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

1. Which opening best corrects the earlier message?

- A** The customer report is withdrawn because we lack a cause.
- B** Our email overstated the cause; humidity remains unconfirmed.
- C** As previously confirmed, humidity is probably responsible.
- D** We have slightly refined our wording but the conclusion is unchanged.

2. Which deadline statement is supported?

- A** I will send an investigation update by 10 a.m. on October 2.
- B** The unit will be repaired by 10 a.m. on October 2.
- C** We guarantee no further interruptions before October 2.
- D** The investigation must be closed before the next message.

3. Which request could help distinguish causes without presuming humidity is responsible?

- A** Supply only a summary supporting the first email's conclusion.
- B** Provide event times, installed configuration, and available environmental records linked to L18.
- C** Confirm that each interruption was humidity-related before sending the original logs.
- D** Remove events that occurred during apparently normal humidity.

4. What should the correction not imply?

- A** That the investigation is still open
- B** Permission to change operation or perform an unapproved repair
- C** That the earlier causal claim was too strong
- D** That the cause remains unresolved

Answers and explanations: [page 97](#).

Correct the cause without losing the customer

E Conversations / turns 1-10

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

WORD + PHRASE BANK

field report / case owner / correction notice / Closure confirmation / service instruction / working hypothesis / resolution commitment / causal statement / traceable record / interim update

07

- 01 **Nia:** Your email said humidity caused the two interruptions on L18. Our team now understands that the cause has not been confirmed, so we need the record corrected.
- 02 **Marco:** You are right. The earlier ¹ _____ was too strong. Humidity is unconfirmed, and I will explicitly correct the message rather than quietly change the wording in our next update.
- 03 **Nia:** Please keep the two interruptions in the record. Uncertainty about the explanation does not change what our operators reported or the disruption they experienced.
- 04 **Marco:** We will preserve the ² _____. The correction concerns our explanation, not whether your team raised the events. Both reports remain part of the investigation for this unit.
- 05 **Nia:** Should our internal summary remove humidity entirely? I want it to be accurate without hiding an explanation that your engineers are still examining.
- 06 **Marco:** Describe it as a ³ _____. It is one explanation being investigated, not a confirmed finding and not something we have ruled out through completed checks.
- 07 **Nia:** Please identify the earlier email in the correction. People are still forwarding it, so they need an explicit statement that the cause has not been confirmed.
- 08 **Marco:** I will issue a ⁴ _____ on the same case. It will say that our previous message overstated the cause and that the investigation remains open.
- 09 **Nia:** We can supply the event times and the installed version information. Tell us how to connect those records so the engineers can distinguish the two occurrences.
- 10 **Marco:** Each ⁵ _____ should identify L18, the event time, and the installed configuration. We will coordinate the evidence request through the case rather than ask you to guess which material matters.

The conversation, continued

E Conversations / turns 11-20

Continue with the word bank on page 64.

- 11 **Nia:** Planning needs a date. What can you commit to now, and what is still preventing a resolution estimate?
- 12 **Marco:** I cannot give a ⁶ _____ yet. I can commit to an update by ten a.m. on October second, and I understand that uncertainty creates a planning burden for you.
- 13 **Nia:** What will that update contain if the cause is still open? Saying we are working on it would give us little new information.
- 14 **Marco:** The ⁷ _____ will separate confirmed findings, open hypotheses, outstanding evidence, and the next review point. It will not turn the communication deadline into a promised repair date.
- 15 **Nia:** One colleague asked whether the correction means we should change how the unit is used or attempt a repair. We have not received an authorized instruction.
- 16 **Marco:** This message is not a ⁸ _____. Any operating or repair direction must come through the authorized product-specific process. I will route that question to the responsible team rather than improvise an answer.
- 17 **Nia:** Please clarify ownership. We have heard from several people, and we need to know who will collect the questions and respond.
- 18 **Marco:** I will remain the ⁹ _____ for communication. Your questions will stay linked to the same record, and I will identify the appropriate escalation contact if additional attention is needed.
- 19 **Nia:** That gives us a usable correction and a clear next contact. I will circulate the amended status, but I will not mark the technical issue resolved.
- 20 **Marco:** Correct. ¹⁰ _____ comes only when the relevant conditions are met. Today's correction restores an accurate record while the investigation and technical decisions continue.

07

ANSWERS / EXPLANATIONS: PAGE 98

1. causal statement | 2. field report | 3. working hypothesis | 4. correction notice | 5. traceable record | 6. resolution commitment | 7. interim update | 8. service instruction | 9. case owner | 10. Closure confirmation

An update is not a repair promise

F Say it

A supplier wrongly called contamination the confirmed cause. It is still a hypothesis. The investigation is open. The supplier promises a progress report Friday at noon, not a repair.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

open / correction / hypothesis / update

Supplier: "Our earlier claim needs a ¹ _____."

Customer: "Contamination remains a ² _____."

Supplier: "Friday noon is the next ³ _____."

Customer: "The investigation remains ⁴ _____."

ANSWERS / EXPLANATIONS: PAGE 98

1. correction | 2. hypothesis | 3. update | 4. open

Rehearse with a fixed script

1. Read the customer exchange. Explicitly correct the earlier humidity claim while retaining the two reported L18 interruptions.
2. Switch roles. Promise the October 2, 10 a.m. update without creating a repair deadline or operating instruction.
3. Complete and check the transfer. Withdraw confirmed contamination and retain a hypothesis, Friday-noon update, and open investigation.

WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

The same number, a different unit

A Read the situation

Explain an interface mismatch using configuration, units, and affected evidence while coordinating a controlled integration decision.

Hardware lead Isabel and software lead Noah integrate a fictional measurement display with no control output. Approved interface document revision 1 defines each transmitted count as 1 mm. Hardware test build H2 instead sends hundredths of a millimeter: 100 counts means 1 mm. Existing software S1 still interprets 100 counts as 100 mm. Earlier tests used H1, S1, and revision 1. A revision-2 interface proposal is not approved, and the impact review and retest plan are incomplete. No production release is authorized.

PEOPLE / Isabel: Hardware lead | Noah: Software lead

08

COMPATIBILITY INCLUDES MEANING

Teams may use the same connector, packet length, and field name while interpreting a value differently. State the units and scaling explicitly. When an interface changes, connect the proposed contract, actual builds, affected tests, and approval route instead of assuming a successful connection proves compatibility.

Briefing checks

Circle one answer in each item. Use the briefing.

1. What does 100 counts mean for H2?

A 100 mm

B 0.01 mm

C 10,000 mm

D 1 mm

2. What does S1 currently display for those 100 counts?

A 100 mm

B 1 mm

C 0.01 mm

D An explicitly supplied error code

3. Which configuration has the earlier test evidence?

A H2 with an approved revision 2

B Every possible hardware and software pairing

C H2 with a completed retest plan

D H1 with S1 under revision 1

Answers and explanations: page 99.

The vocabulary that does the work

B Find the words / 1 of 2

Read the definitions and say the common word combinations aloud.

interface control document

A controlled definition of an interface, abbreviated ICD.

IN USE update the interface control document

interface contract

The agreed expectations for interaction between components.

IN USE define the interface contract

configuration baseline

The approved set of identified versions used as a reference.

IN USE establish the configuration baseline

hardware build

An identified version or assembly of hardware.

IN USE identify the hardware build

software version

An identified release or build of software.

IN USE record the software version

integration test

A test of components working together.

IN USE run an integration test

regression test

A check that changes have not broken previously working behavior.

IN USE define regression tests

unit scaling

The conversion between encoded values and physical units.

IN USE document unit scaling

payload

The data carried inside a message or transmission.

IN USE inspect the payload

data type

The defined representation and permitted operations for a value.

IN USE specify the data type

field width

The size allocated to a data field.

IN USE confirm the field width

byte order

The sequence used to store or transmit a multi-byte value.

IN USE specify byte order

Precision check

The mismatch is semantic: the number has a different unit scale. For H2, 100 counts times 0.01 mm per count equals 1 mm. S1 applies 1 mm per count and displays 100 mm. A successful message transfer would not establish correct interpretation.

The vocabulary that does the work

B Find the words / 2 of 2

Read the definitions and say the common word combinations aloud.

message schema

The defined structure and meaning of a message.

IN USE version the message schema

backward compatibility

Ability to work with an earlier version under stated conditions.

IN USE assess backward compatibility

version negotiation

A process for identifying or agreeing compatible interface versions.

IN USE define version negotiation

configuration mismatch

A combination of versions that does not share the expected contract.

IN USE identify a configuration mismatch

change impact analysis

Assessment of the consequences of a proposed or actual change.

IN USE complete change impact analysis

affected test

A test whose relevance or expected result changes with the modification.

IN USE identify affected tests

test evidence reuse

Using earlier results where their continued applicability is justified.

IN USE justify test evidence reuse

conversion factor

A multiplier used to express a value on another unit basis.

IN USE apply the conversion factor

interface owner

The role responsible for maintaining the interface definition.

IN USE identify the interface owner

change approval

Authorization to make a defined controlled change.

IN USE obtain change approval

configuration record

A record identifying the versions and settings used.

IN USE preserve the configuration record

integration readiness

The state of having the agreed inputs and evidence needed to integrate.

IN USE assess integration readiness

08

Precision check

Earlier test evidence still describes H1 with S1 under revision 1. It does not automatically validate H2, but not every unrelated result must automatically be discarded. Reuse needs an impact-based justification and a recorded configuration.

Phrases you can actually use

C Notice the language / 1 of 2

IDENTIFY THE PAIRING

"Which hardware build and software version are under test?"

NAME THE CONTRACT

"The approved interface is still revision 1."

STATE THE CHANGE

"H2 sends hundredths of a millimeter."

SHOW THE CONSEQUENCE

"One hundred counts means 1 mm for H2 but 100 mm for S1."

SEPARATE CONNECTION FROM MEANING

"A received message can still be interpreted incorrectly."

CHECK THE SCALE

"What conversion factor does this version apply?"

PRESERVE THE OLD RESULT

"The earlier evidence applies to H1 with S1."

AVOID BLANKET REUSE

"That result does not automatically cover the new pairing."

Compatible

Includes shared meaning, not only physical connection.

Count

Has no useful physical meaning without the scaling definition.

Latest

An ambiguous version description in a technical record.

Phrases you can actually use

C Notice the language / 2 of 2

AVOID BLANKET DELETION

"Unaffected evidence may be reusable with justification."

BOUND THE REVIEW

"Assess decoding, display, records, and affected tests."

REQUIRE A CONTROLLED DECISION

"Route the proposed change through the approval process."

RECORD THE EXACT CONFIGURATION

"Store the hardware, software, and interface versions with the result."

IDENTIFY THE PROPOSAL

"Revision 2 is not yet approved."

NAME INTERFACE OWNERSHIP

"Who owns the shared interface definition?"

PLAN THE RETEST

"Link each affected requirement to the appropriate test."

CLOSE WITHOUT OVERCLAIM

"Integration and production release remain pending."

08

Passed before

Needs the exact configuration and test scope.

Reuse

Requires continued applicability, not convenience alone.

Approved change

Distinct from a test build or draft document already existing.

Choose the precise message

D Check your understanding

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

1. Which diagnosis matches the facts?

- A** The earlier test proves S1 handles every scaling convention.
- B** H2 and S1 use different scaling for the same transmitted count.
- C** The customer has approved revision 2.
- D** The interface is compatible because a number can be transmitted.

08

2. For H2's 100-count message, which correction addresses the unit meaning?

- A** Keep the displayed value 100 and relabel it mm, because the packet arrived.
- B** Multiply 100 by 100 to compensate for the smaller encoded unit.
- C** Multiply 100 by 0.01 mm per count to obtain 1 mm; changing only the display label is insufficient.
- D** Divide 100 by 0.01 to obtain 10,000 mm.

3. How should earlier evidence be treated?

- A** Preserve it and assess which results remain applicable to the changed configuration.
- B** Relabel all earlier results as H2 tests.
- C** Delete all results without an impact review.
- D** Declare the new interface approved because earlier tests passed.

4. What is needed before claiming integration readiness?

- A** Only a message received without a connection error
- B** Only the existence of a draft revision 2
- C** A statement that both teams prefer the new units
- D** A resolved interface definition, controlled configuration, and appropriate reviewed test evidence

Answers and explanations: page 99.

The same number, a different unit

E Conversations / turns 1-10

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

WORD + PHRASE BANK

conversion factor / regression tests / integration readiness / change impact analysis / test evidence reuse / unit scaling / configuration record / configuration mismatch / interface control document / interface owner

08

- 01 **Noah:** The message arrives without an error, but S1 displays one hundred millimeters. You expected one millimeter. Can we compare the transmitted count and its meaning?
- 02 **Isabel:** H2 sends one hundred, but its ¹ _____ is now hundredths of a millimeter. The approved revision-one interface still defines whole millimeters per count.
- 03 **Noah:** Then S1 is applying its documented multiplier of one. The number was received, but this pairing interprets it one hundred times too large.
- 04 **Isabel:** Yes, we have a ² _____. A successful packet transfer does not demonstrate agreement about physical units.
- 05 **Noah:** Our earlier test passed. Before anybody uses that result to clear H2, which actual builds and interface revision did the test cover?
- 06 **Isabel:** The ³ _____ says H1, S1, revision one. Preserve that result as evidence for that combination; it was not a test of H2.
- 07 **Noah:** I found the revision-two proposal in the folder. Its presence does not mean the new scale is approved, even though the hardware test build already uses it.
- 08 **Isabel:** Correct. The ⁴ _____ still needs controlled review and approval. A new test build cannot silently replace the shared definition for the software team.
- 09 **Noah:** For the proposed scale, should I divide one hundred by zero point zero one? That would produce ten thousand millimeters, which makes the mismatch worse.
- 10 **Isabel:** Multiply by the ⁵ _____ instead: zero point zero one millimeters per count. One hundred counts then gives one millimeter. Include that worked example in the proposal.

The conversation, continued

E Conversations / turns 11-20

Continue with the word bank on page 73.

- 11 **Noah:** Could we just change the label beside the displayed number? The field name and packet size are unchanged, so the patch could look quite small.
- 12 **Isabel:** A label alone would leave the value wrong. The ⁶ _____ must follow decoding, display, stored records, and any other uses of the physical quantity.
- 13 **Noah:** Should I discard every earlier test, or can some results remain useful? I want to avoid both unsupported reuse and rerunning unrelated checks without a reason.
- 14 **Isabel:** Assess ⁷ _____ against the change. Keep applicable results with their original configuration references, and identify where changed behavior needs new evidence.
- 15 **Noah:** I will list affected requirements and expected values for the proposed pairing. The tests should catch both a wrong factor and a correct-looking number from the wrong configuration.
- 16 **Isabel:** Include ⁸ _____ for previously working behavior within the assessed scope. A direct scaling check alone does not establish that the rest of the affected behavior is unchanged.
- 17 **Noah:** Who will maintain the shared definition? Separate hardware and software notes could leave us with this disagreement again at the next test.
- 18 **Isabel:** We need a named ⁹ _____ and a confirmed approval route. I will bring the hardware-build details; please attach the software interpretation and affected-test list.
- 19 **Noah:** I will. Today's outcome is an understood scaling mismatch and assigned evidence preparation, not an approved new interface or production release.
- 20 **Isabel:** Agreed. Establishing ¹⁰ _____ still needs the controlled definition, compatible configuration, and reviewed evidence. We will not relabel the earlier pass as proof for H2.

08

ANSWERS / EXPLANATIONS: PAGE 100

1. unit scaling | 2. configuration mismatch | 3. configuration record | 4. interface control document | 5. conversion factor | 6. change impact analysis | 7. test evidence reuse | 8. regression tests | 9. interface owner | 10. integration readiness

A count needs its conversion

F Say it

A proposed interface uses 0.1 mm per count. A test message contains 50 counts, so its intended distance is 5 mm. Old software uses 1 mm per count and displays 50 mm. The proposal is unapproved.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

5 / 50 / approval / scaling

08

Engineer: "The proposed distance is ¹ _____ mm."

Reviewer: "The old software displays ² _____ mm."

Engineer: "The difference comes from unit ³ _____."

Reviewer: "The proposed change still needs ⁴ _____."

ANSWERS / EXPLANATIONS: PAGE 100

1. 5 | 2. 50 | 3. scaling | 4. approval

Rehearse with a fixed script

1. Read turns 1-10. Contrast H2's 100 counts as 1 mm with S1's 100 mm display; correct division to multiplication by 0.01.
2. Switch roles for turns 11-20. Keep evidence for H1/S1/revision 1 separate from the unapproved H2 proposal.
3. Complete and check the transfer. Contrast 5 mm intended with 50 mm displayed, then state that approval remains outstanding.

WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

A smoother plot is not a convergence study

E More workplace conversations / 1-10

Challenge a simulation claim by asking about the quantity of interest, model assumptions, and supporting evidence.

Three finite-element runs use progressively finer meshes. Displacement changes little, but peak stress at an idealized sharp corner keeps increasing. The design lead wants a green pass label. No acceptance threshold or release conclusion is supplied.

PEOPLE / Yara: Structural analyst | Finn: Design lead

Use each bank entry once, without changing its form. Answers follow turn 20.

09

WORD + PHRASE BANK

quantity of interest / boundary conditions / Mesh convergence / Verification / sensitivity analysis / stress singularity

- 01 **Finn:** The latest plot looks much smoother. Can we call the model converged and put the result into the release review?
- 02 **Yara:** We need to name the ¹ _____. Displacement changes little between these runs, but the peak stress keeps rising.
- 03 **Finn:** I assumed a finer mesh improved every result in the same way. Does the stable displacement not settle the whole model?
- 04 **Yara:** No. ² _____ concerns the relevant output and purpose. One stable measure does not establish stability for every value.
- 05 **Finn:** The rising stress is at that sharp inside corner. Could the model geometry itself explain why the local maximum does not settle?
- 06 **Yara:** A ³ _____ is one possibility to investigate. We must assess the idealization rather than automatically interpret each higher peak as a new physical measurement.
- 07 **Finn:** But we should not simply delete the red area from the plot and call the design satisfactory either.
- 08 **Yara:** Exactly. We need a justified assessment of the actual feature and intended failure criterion, with the limitations visible. Changing the display is not solving the engineering question.
- 09 **Finn:** What should the review package show about the supports and loads? The slide currently only has the colored result.
- 10 **Yara:** The ⁴ _____, load case, material assumptions, geometry, and relevant solution settings. The picture alone does not tell reviewers what physical problem we solved.

The conversation, continued

E More workplace conversations / 11-20

Continue with the word bank on page 76.

- 11 **Finn:** The supplier says their model uses the same material name. Should we expect identical results when we compare them?
- 12 **Yara:** Not without checking the actual properties, formulations, geometry, and conditions. A shared label does not establish equivalent inputs or methods.
- 13 **Finn:** We have also checked the implementation against a known calculation. Is that the same as showing that the model represents the real assembly?
- 14 **Yara:** No. ⁵ _____ and comparison with relevant physical evidence answer different questions. Passing a numerical check does not automatically validate the intended real-world use.
- 15 **Finn:** Then the hardware comparison still matters. We need to say which measurements and conditions support the use we are proposing.
- 16 **Yara:** Yes, and include ⁶ _____ where appropriate. A conclusion that changes sharply with an uncertain input needs that sensitivity explained.
- 17 **Finn:** For today, I will remove the blanket green label. We can report stable displacement in these runs and the unresolved local-stress interpretation separately.
- 18 **Yara:** That is accurate. I will provide the refinement results and the model assumptions for review, without treating the visual smoothness as acceptance evidence.
- 19 **Finn:** Please also identify what further work is needed for the release question. The reviewers need an actionable gap, not just a warning that models have limitations.
- 20 **Yara:** I will link each open question to the intended claim. That lets us decide what the analysis can support and what still requires assessment.

09

ANSWERS / EXPLANATIONS: PAGE 101

1. quantity of interest | 2. Mesh convergence | 3. stress singularity | 4. boundary conditions | 5. Verification | 6. sensitivity analysis

One stable output is used to approve every output

F Say it in a new situation

A thermal model gives a stable average temperature under mesh refinement, but its local peak is still changing materially.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

report / peak / assessment / criterion

09

Lead: Stability of the average does not establish the local ¹ _____.

Analyst: State the specific quantity of interest for the ² _____.

Lead: Keep the remaining numerical limitation visible in the ³ _____.

Analyst: A smooth image is not an acceptance ⁴ _____.

ANSWERS / EXPLANATIONS: PAGE 101

1. peak | 2. assessment | 3. report | 4. criterion

Read, switch, repeat

1. Check the four answers and read the complete exchange aloud.

2. Switch roles. Read it again with the word bank covered.

3. Return to the extended conversation. Read turns 1-10, then switch roles for turns 11-20. Use the corrected expressions.

SOLO PRACTICE

Read both roles. Pause briefly at each change of speaker. Repeat any line that needed an answer check.

The replacement fits, but is it an approved alternate?

E More workplace conversations / 1-10

Evaluate a proposed component substitution across requirements and product configuration.

A proposed replacement sensor has the same connector and envelope dimensions as the unavailable original. Its output scale, accuracy over temperature, and environmental rating remain unassessed. The bill of materials lists only the original. A buyer asks about production use.

PEOPLE / Milo: Buyer | Serena: Design engineer

Use each bank entry once, without changing its form. Answers follow turn 20.

10

WORD + PHRASE BANK

interface requirements / approved alternate / bill of materials / transfer function / effectivity / form-and-fit

- 01 **Milo:** The usual sensor is unavailable. This one fits the same space and connector, and the supplier calls it a drop-in replacement. Can I order it for production?
- 02 **Serena:** Send me the exact part number and datasheet. A matching envelope establishes part of the ¹ _____ comparison, not the complete engineering equivalence.
- 03 **Milo:** The connector looks identical in the photograph. I have not compared the pin assignments or electrical characteristics yet.
- 04 **Serena:** We need the actual ² _____, including those details. A photograph cannot establish that two connected parts behave compatibly.
- 05 **Milo:** What about the output? The supplier lists the same measured quantity, but the scale is on a different page of the datasheet.
- 06 **Serena:** Compare the ³ _____. Measuring the same quantity does not mean the receiving system can interpret the output without a change.
- 07 **Milo:** I also see an accuracy figure measured at room temperature. Our existing requirement covers a wider operating range.
- 08 **Serena:** Then we need evidence across the applicable conditions. A favorable room-temperature value does not automatically meet an accuracy-over-temperature requirement.
- 09 **Milo:** The supplier says they sell it to several similar customers. I can ask for supporting information, but that is not our product assessment.
- 10 **Serena:** Right. We should compare our requirements, including environmental and reliability needs, and identify the gaps. Other customers may have different configurations and use conditions.

The conversation, continued

E More workplace conversations / 11-20

Continue with the word bank on page 79.

- 11 **Milo:** Could I add the part to the purchasing list while engineering finishes the comparison? That would make it easier to place the order quickly later.
- 12 **Serena:** Keep it as a candidate. The controlled ⁴ _____ must not show it as a production-approved part before the required decision.
- 13 **Milo:** Understood. Any samples for evaluation would need their own purchase purpose and handling so they are not mistaken for production stock.
- 14 **Serena:** Exactly. An ⁵ _____ has a specific assessed use. We should not let a convenient stock code imply approval across every product.
- 15 **Milo:** If the comparison supports a change, do we also need to identify which units or builds can use it?
- 16 **Serena:** Yes. The ⁶ _____ needs to be clear, along with any associated documentation, software, test, or service implications.
- 17 **Milo:** I will give you the shortage timing, exact candidate details, and what the supplier can provide. I will not promise production availability yet.
- 18 **Serena:** I will return a requirement-by-requirement comparison and the required review steps. We can show which checks are complete and which still block a decision.
- 19 **Milo:** That gives planning a real status: candidate found, equivalence under assessment, no production substitution approved.
- 20 **Serena:** Yes. The goal is a workable alternative with a documented basis, not merely a part that can be plugged into the same connector.

10

ANSWERS / EXPLANATIONS: PAGE 102

1. form-and-fit | 2. interface requirements | 3. transfer function | 4. bill of materials | 5. approved alternate | 6. effectivity

A sample order becomes an apparent production approval

F Say it in a new situation

A buyer orders a candidate component for engineering evaluation. The purchasing record mistakenly labels it an approved production alternate.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

alternates / requirements / history / use

10

Engineer: These parts were ordered for evaluation, not production

¹ _____.

Buyer: Correct the status without losing the purchase ² _____.

Engineer: Complete the comparison against the applicable ³ _____.

Buyer: Keep the candidate separate from approved ⁴ _____.

ANSWERS / EXPLANATIONS: PAGE 102

1. use | 2. history | 3. requirements | 4. alternates

Read, switch, repeat

1. Check the four answers and read the complete exchange aloud.
2. Switch roles. Read it again with the word bank covered.
3. Return to the extended conversation. Read turns 1-10, then switch roles for turns 11-20. Use the corrected expressions.

SOLO PRACTICE

Read both roles. Pause briefly at each change of speaker. Repeat any line that needed an answer check.

Two settings changed, so which one improved the result?

E More workplace conversations / 1-10

Negotiate an experiment that separates factors and measures variation without overstating a pilot result.

One bench run uses fan setting A and vent X; another uses fan B and vent Y. The second records a lower temperature, but ambient conditions also differ. The team discusses a follow-up design, not authorized test settings or a release decision.

PEOPLE / Harper: Engineering lead | Dae: Test engineer

Use each bank entry once, without changing its form. Answers follow turn 20.

WORD + PHRASE BANK

interaction / randomization / design matrix / response variable / factors / replicates

11

- 01 **Harper:** The second prototype ran cooler. Can we say vent design Y is responsible and recommend it to the design review?
- 02 **Dae:** Not from these two runs. We changed two ¹ _____ together, and the ambient temperature was different as well.
- 03 **Harper:** Then the fan setting could explain some of the difference. We have not tested A with Y or B with X.
- 04 **Dae:** Exactly. The ² _____ should show the combinations needed to address our question, rather than only the two configurations that happened to be available.
- 05 **Harper:** Would looking at both factors together help us see whether the effect of the vent depends on the fan setting?
- 06 **Dae:** Yes. That is an ³ _____ question. We should choose a reviewed design capable of examining it instead of assuming the two effects simply add together.
- 07 **Harper:** We also need to define cooler. Are we comparing the same sensor location, the same elapsed point, and the same summary of the temperature data?
- 08 **Dae:** We need a precise ⁴ _____. Otherwise the runs might use different summaries while the slide presents them as one measure.
- 09 **Harper:** The team suggests taking several readings from the same run and treating them as several independent tests. That would give us a larger count.
- 10 **Dae:** Repeated readings help describe that run, but they are not automatically independent ⁵ _____. The plan must distinguish within-run readings from independently repeated runs.

The conversation, continued

E More workplace conversations / 11-20

Continue with the word bank on page 82.

- 11 **Harper:** How should we handle changing ambient conditions? We cannot assume the laboratory stays identical throughout the work.
- 12 **Dae:** Record the relevant conditions and consider them in the design. Depending on the constraints, blocking may help account for known variation rather than let it follow one design option systematically.
- 13 **Harper:** Could we run all X configurations first and all Y configurations afterward to make setup easier?
- 14 **Dae:** That may confound design with time. Consider appropriate ⁶ _____ or a justified restricted design, while preserving the applicable safety and test controls.
- 15 **Harper:** I will not prescribe the run order in the review meeting. Please bring a plan showing the comparisons, repeated runs, and practical constraints.
- 16 **Dae:** I will also show what conclusions the proposed design can and cannot support. A limited pilot may identify a useful direction without establishing every operating condition.
- 17 **Harper:** For the existing result, we can say the B-and-Y run recorded a lower temperature under its stated conditions. We cannot assign that difference solely to the vent.
- 18 **Dae:** Correct. Preserve both runs and the ambient data. The first comparison is informative, but its limitations need to travel with the result.
- 19 **Harper:** Once the plan is reviewed, we can decide the work and resources. Until then, this discussion should not be read as permission to change the bench procedure.
- 20 **Dae:** Agreed. We are defining a better question and evidence plan, not converting an uncontrolled comparison into an approved design choice.

11

ANSWERS / EXPLANATIONS: PAGE 103

1. factors | 2. design matrix | 3. interaction | 4. response variable | 5. replicates | 6. randomization

Repeated readings are counted as independent trials

F Say it in a new situation

One prototype undergoes one test run. Ten sensor readings from that run are described as ten independent trials.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

count / variation / supports / trials

Lead: Ten readings from one run are not automatically ten independent

¹ _____.

Engineer: Identify the experimental unit and the source of ² _____.

Lead: Keep the measurement count distinct from the run ³ _____.

Engineer: Limit the conclusion to what the design actually ⁴ _____.

11

ANSWERS / EXPLANATIONS: PAGE 103

1. trials | 2. variation | 3. count | 4. supports

Read, switch, repeat

1. Check the four answers and read the complete exchange aloud.
2. Switch roles. Read it again with the word bank covered.
3. Return to the extended conversation. Read turns 1-10, then switch roles for turns 11-20. Use the corrected expressions.

SOLO PRACTICE

Read both roles. Pause briefly at each change of speaker. Repeat any line that needed an answer check.

01 Briefing and language checks

Requirements and Constraints

A Briefing checks / page 4

1. What is included in the proposed mass limit?

B. The empty enclosure, cover, and fasteners

The briefing explicitly includes the enclosure, cover, and fasteners while excluding electronics.

2. What is the status of five years?

D. A proposed service-life target

Five years is proposed while conditions and the verification method remain unresolved.

3. Which condition is still open?

A. Operating temperature

Operating temperature is listed among the unresolved conditions needed to define the proposal.

D Language checks / page 9

1. Which statement accurately records the mass proposal?

D. Maximum 1.2 kg for the empty enclosure including cover and fasteners, excluding electronics

The complete boundary prevents different teams from weighing different assemblies against the same number.

2. Which sentence avoids overstating durability?

A. Five years is the proposed service life; operating conditions and verification remain open.

The proposed duration is not evidence of achieved life under unspecified conditions.

3. What should happen to unresolved environmental assumptions?

C. Record them with owners for confirmation.

Recording ownership preserves uncertainty and provides a route to resolve the missing inputs.

4. A prototype meets the proposed mass boundary at 1.15 kg. What does that establish about the five-year target?

C. Nothing by itself; a mass result does not demonstrate service life under undefined conditions.

Mass and durability are different properties with separate boundaries and evidence needs. The mass comparison does not resolve life, conditions, or approval.

01 Conversation answers

Requirements and Constraints

E Dialogue gaps

Pages 10-11

1. maximum mass. Turn 2. Maximum mass states the upper limit, unlike an undefined preference for a lightweight product.

2. system boundary. Turn 4. The system boundary identifies which physical items count toward the proposed mass limit.

3. service life. Turn 6. Service life is the intended period of use under specified conditions, which are still incomplete here.

4. operating envelope. Turn 8. The operating envelope defines the conditions under which the design is expected to operate.

5. assumption register. Turn 10. An assumption register preserves unconfirmed premises and their ownership until they are resolved.

6. verification method. Turn 12. The verification method specifies how evidence will be obtained to check the requirement.

7. rationale. Turn 14. The rationale records why the requirement exists and supports later review of changes.

8. traceability. Turn 16. Traceability links the customer need, requirement, and later evidence without pretending unfinished items are complete.

9. baseline. Turn 18. A baseline is an approved reference, not merely the latest proposal discussed at a meeting.

10. requirement owner. Turn 20. The requirement owner is responsible for maintaining the requirement and coordinating resolution of its open details.

F Transfer conversation

Page 12

1. included

Engineer: "The lid and screws are included."

The supplied boundary explicitly counts the housing, lid, and screws.

2. excluded

Customer: "The batteries are excluded."

Batteries are explicitly outside the proposed mass boundary in this case.

3. unresolved

Engineer: "Humidity remains unresolved."

The briefing states that operating humidity has not yet been agreed.

4. baseline

Customer: "This is a proposal, not an approved baseline."

Approval is pending, so the proposal is not an approved reference.

02 Briefing and language checks

Design Reviews and Technical Pushback

A Briefing checks / page 13

1. What does the CAD model currently show?

C. 2 mm nominal clearance

The model supplies nominal geometry but not the missing tolerance evidence.

2. What does R-14 require?

A. At least 1 mm across approved dimensional tolerances

The stated requirement applies across tolerances, not only at nominal dimensions.

3. What is not established?

A. Whether the minimum clearance meets R-14

Without the stack-up and confirmed revision, compliance with the minimum remains unverified.

D Language checks / page 18

1. Which review comment is best supported?

D. R-14 remains unverified pending the stack-up and revision confirmation.

This comment identifies the evidence gap without converting uncertainty into failure or compliance.

2. Which closure action addresses both missing inputs for R-14?

A. Confirm the supplier drawing revision and review the interface tolerance stack-up against the 1 mm minimum.

The open item concerns combined allowed variation on an identified configuration. Repeating nominal geometry, checking a filename alone, or recording attendance does not resolve it.

3. What may the minutes record now?

C. An open verification item and its assigned follow-up

The current record supports a pending item, not an invented approval, failure, or agreement.

4. What would justify closing the item?

D. Reviewed evidence for the confirmed configuration against R-14

Closure must resolve the specific requirement and configuration gap identified in the review.

02 Conversation answers

Design Reviews and Technical Pushback

E Dialogue gaps

Pages 19-20

- 1. tolerance stack-up.** Turn 2. The tolerance stack-up evaluates combined dimensional variation rather than relying on the nominal CAD gap.
- 2. Nominal clearance.** Turn 4. Nominal clearance describes the reference geometry; it does not establish the smallest gap across allowed variation.
- 3. drawing revision.** Turn 6. The drawing revision identifies the supplier geometry and tolerances that the analysis must actually evaluate.
- 4. review action.** Turn 8. A review action assigns a specific task arising from the meeting, rather than leaving an unowned concern.
- 5. closure evidence.** Turn 10. Closure evidence must show that the specific requirement gap has been resolved on the confirmed configuration.
- 6. Conditional approval.** Turn 12. Conditional approval must follow the applicable process and state its limits; it cannot silently mean unrestricted release.
- 7. release authority.** Turn 14. The release authority is the role permitted to make the release decision, distinct from individual reviewers.
- 8. technical objection.** Turn 16. A technical objection challenges a defined claim with reasons and a route to resolution, not a colleague's competence.
- 9. open item.** Turn 18. An open item records the unresolved matter until the stated closure conditions are satisfied.
- 10. Action closure.** Turn 20. Action closure requires verified resolution of the assigned task, not simply an intention to provide evidence later.

F Transfer conversation

Page 21

1. nominal

Reviewer: "Three millimeters is the nominal clearance."

The value comes from reference geometry, not a completed variation analysis.

2. confirmation

Engineer: "The supplier revision needs confirmation."

The briefing explicitly says the supplier revision has not been confirmed.

3. unverified

Reviewer: "The minimum remains unverified."

A missing stack-up prevents verification of the required minimum across tolerances.

4. evidence

Engineer: "The item closes with reviewed evidence."

Closure needs evidence addressing the requirement, not merely a repeated nominal value.

03 Briefing and language checks

Failure Modes and Reliability

03

A Briefing checks / page 22

1. What was directly observed?

D. S7 stopped producing an output at minute 46.

The briefing gives an output interruption on one identified sample, not a confirmed explanation.

2. Which statement accurately describes the cause?

B. Connector damage is an unconfirmed hypothesis.

Neither reproduction nor setup checks have established the cause of the event.

3. What cannot be calculated from the supplied facts alone?

C. A representative population failure rate

A single reported event without an adequate exposure and sampling basis does not establish population reliability.

D Language checks / page 27

1. Which draft sentence should replace "connector damage caused the failure"?

B. S7 lost output at minute 46; connector damage remains a hypothesis.

The replacement preserves the observation while accurately qualifying the unconfirmed cause.

2. Why preserve the event log and sample identity?

B. They connect the investigation to the actual event and configuration.

Traceable records let investigators evaluate the event without inventing a causal conclusion.

3. A later repeat run shows no interruption. What should happen to the original S7 result?

C. Retain the minute-46 event and investigate why the observations differ; one repeat does not identify the cause.

A non-recurrence in a later run does not erase the original observation or distinguish the connector, fixture, setup, and other possible explanations.

4. Which wording distinguishes containment from corrective action?

B. Exposure controls may begin while the cause-based action remains undecided.

Immediate exposure management and action against an established cause have different purposes.

03 Conversation answers

Failure Modes and Reliability

03

E Dialogue gaps

Pages 28-29

1. failure mode. Turn 2. The failure mode is loss of output, the observed way the sample stopped meeting its intended function.

2. hypothesis. Turn 4. A hypothesis is a plausible explanation to investigate, not an established finding based on this event.

3. test setup. Turn 6. The test setup includes the fixture and connections whose condition may affect interpretation of the sample result.

4. event log. Turn 8. The event log retains time-ordered evidence connecting the observed interruption to the test and sample.

5. residual uncertainty. Turn 10. Residual uncertainty includes the unresolved cause and the limits on any population-level reliability claim.

6. failure effect. Turn 12. The failure effect is the consequence of the mode at a stated level, distinct from the mechanism causing it.

7. FMEA. Turn 14. FMEA connects possible failure modes, their effects, causes, and controls without converting hypotheses into proven causes.

8. containment. Turn 16. Containment limits immediate exposure while the underlying cause and permanent response may still be unresolved.

9. corrective action. Turn 18. Corrective action addresses an established cause and requires appropriate evidence that the intended correction works.

10. failure analysis. Turn 20. Failure analysis systematically evaluates the preserved evidence and possible explanations rather than endorsing the first plausible story.

F Transfer conversation

Page 30

1. mode

Tester: "Loss of signal is the observed failure mode."

The mode describes how the required function failed, not why it failed.

2. hypothesis

Engineer: "Cable damage is still a hypothesis."

The suspected explanation has not been confirmed by the stated checks.

3. log

Tester: "We will preserve the original log."

The available original log is relevant evidence for the recorded event.

4. unresolved

Engineer: "The root cause remains unresolved."

Neither the cable nor the fixture has been checked, so the cause is not established.

04 Briefing and language checks

Testing, Validation, and Data Interpretation

04

A Briefing checks / page 31

1. Which result is supported?

C. All three prototypes met the stated bench criterion at 20 degrees Celsius.

Each recorded maximum is below 200 milliseconds, but the conditions and sample are limited.

2. Which value was the largest recorded maximum?

C. 175 milliseconds

The three recorded maxima are 150, 160, and 175 milliseconds.

3. What remains untested in this case?

C. Performance at the other temperatures in the intended range

Only 20 degrees Celsius was tested; the remaining intended temperatures were not covered.

D Language checks / page 36

1. Which presentation headline is most accurate?

D. Three revision-B prototypes met the response-time criterion in the stated bench test.

The headline reports the demonstrated sample result without extending it to untested conditions or units.

2. What does the observed 100% pass fraction mean?

A. All three tested items met this criterion in this test.

A sample pass fraction describes these observations and does not establish universal reliability.

3. The largest recorded maximum is 175 ms against a 200 ms limit. Which statement is precise?

A. The observed gap to the limit is 25 ms; no supplied uncertainty budget establishes a guaranteed margin for all units.

Subtraction gives 25 milliseconds for the reported result. That arithmetic does not extend the sample, cover other temperatures, or supply a measurement-uncertainty assessment.

4. Which limitation is specific and useful?

C. Performance at the other intended temperatures was not tested.

This identifies the missing coverage without denying the observed passes or inventing failures.

04 Conversation answers

Testing, Validation, and Data Interpretation

E Dialogue gaps

Pages 37-38

1. pass fraction. Turn 2. The pass fraction is three passing prototypes out of three tested, limited to this criterion and test.

2. test conditions. Turn 4. Test conditions specify the temperature, duration, and procedure under which the observations were obtained.

3. observed maximum. Turn 6. The observed maximum is the largest recorded value, not a guaranteed limit for every future unit or condition.

4. sample size. Turn 8. Sample size identifies the three prototypes behind the percentage and prevents an impression of broader evidence.

5. test coverage. Turn 10. Test coverage describes which requirements and conditions were actually addressed, exposing the untested temperature range.

6. verification. Turn 12. Verification checks a specified requirement; this bench result does not by itself establish complete intended-use suitability.

7. validation. Turn 14. Validation addresses intended-use needs, which are broader than the single bench criterion described here.

8. extrapolation. Turn 16. Extrapolation extends conclusions beyond the observed temperature; no supporting model or additional evidence is supplied here.

9. reliability claim. Turn 18. A reliability claim requires a defined performance, time, conditions, and supporting basis beyond a small unqualified pass count.

10. evidence boundary. Turn 20. The evidence boundary separates the supported bench result from broader claims about other conditions, units, or lifetime.

F Transfer conversation

Page 39

1. four

Engineer: "The tested sample size was four."

The briefing identifies four prototypes in this particular bench test.

2. hours

Reviewer: "The duration was three hours."

The supplied test duration is three hours, not a service-life result.

3. untested

Engineer: "The other temperatures remain untested."

The briefing explicitly excludes testing at the other intended temperatures.

4. reliability

Reviewer: "The result does not establish population reliability."

No population reliability estimate or adequate supporting basis is supplied.

05 Briefing and language checks

Manufacturability and Cost Engineering

A Briefing checks / page 40

05

1. What is the recurring material-plus-assembly cost of A?

D. \$14 per unit

Adding A's stated material cost of eight dollars and assembly cost of six gives fourteen.

2. What is the total stated cost of B for 3,000 units?

D. \$41,000

Three thousand units at twelve dollars cost thirty-six thousand, plus five thousand in tooling.

3. What remains unresolved after the calculation?

D. Technical suitability, capacity, and approval

The briefing explicitly leaves technical and supply decisions open despite the available cost figures.

D Language checks / page 45

1. Which recommendation is supported at 3,000 units?

B. B has the lower stated cost, subject to unresolved technical and capacity reviews.

B's total is two thousand dollars lower, but cost arithmetic does not resolve the other reviews.

2. At which volume are the stated totals equal?

C. 2,000 units

B's additional four-thousand-dollar tooling cost is offset by two dollars per unit at two thousand units.

3. B saves \$2 per unit for 3,000 units. Why is the stated total saving \$2,000 rather than \$6,000?

A. The \$6,000 recurring saving is offset by B's \$4,000 additional tooling.

Recurring savings are 3,000 times \$2. Tooling differs by \$5,000 minus \$1,000, so the net difference is \$6,000 minus \$4,000 = \$2,000.

4. Why should the model state no scrap?

B. Different production yields could change the cost comparison.

A stated simplification is needed because actual losses may alter the cost of acceptable output.

05 Conversation answers

Manufacturability and Cost Engineering

E Dialogue gaps

Pages 46-47

1. cost model. Turn 2. The cost model combines the same cost categories and quantity for both alternatives rather than comparing material alone.

2. recurring cost. Turn 4. Recurring cost is the material-plus-assembly amount incurred for each unit, excluding the one-time tooling.

3. tooling investment. Turn 6. Tooling investment is the one-time amount added to the recurring production total for each option.

4. production volume. Turn 8. Production volume is the three-thousand-unit basis that determines how recurring savings compare with fixed tooling differences.

5. break-even volume. Turn 10. Break-even volume is two thousand units because the two-dollar saving offsets four thousand dollars of additional tooling.

6. cost trade-off. Turn 12. The cost trade-off exchanges higher upfront tooling for lower recurring assembly-inclusive cost.

7. manufacturability. Turn 14. Manufacturability concerns producing the design by the intended process; a lower calculated price does not establish it.

8. process capability. Turn 16. Process capability concerns meeting specified limits in a stable process and is distinct from a quoted cost or available capacity.

9. yield assumption. Turn 18. The yield assumption is the no-scrap premise used here; different actual yields could change acceptable-unit cost.

10. unit cost. Turn 20. Unit cost must identify included categories and volume; otherwise recurring-only and tooling-inclusive figures can be confused.

F Transfer conversation

Page 48

1. 11000

Engineer: "C totals 11000 dollars."

Nine dollars times one thousand units plus two thousand tooling equals eleven thousand.

2. 13000

Reviewer: "D totals 13000 dollars."

Seven dollars times one thousand units plus six thousand tooling equals thirteen thousand.

3. C

Engineer: "At this quantity, the lower-cost option is C."

C costs eleven thousand, which is two thousand less than D.

4. 2000

Reviewer: "Equal totals occur at 2000 units."

D's four-thousand-dollar additional tooling is offset by two dollars per unit at two thousand.

05

06 Briefing and language checks

Safety Factors and Compliance

A Briefing checks / page 49

1. What does 2.0 describe in the note?

A. A preliminary static-model ratio under its stated convention

The ratio belongs to the preliminary static model and does not establish the broader claims.

2. Which behavior is outside the model's assessment?

C. Fatigue and dynamic-loading effects

The briefing explicitly places fatigue and dynamic loading outside the model.

3. What is missing from the compliance claim?

A. The applicable requirement and acceptance threshold

A compliance conclusion needs an identified basis, which the note currently does not provide.

D Language checks / page 54

1. Which replacement note is justified?

D. The preliminary static model reports 2.0; applicable acceptance requirements and review remain unresolved.

The replacement preserves the numerical result and clearly identifies the missing decision basis.

2. Why is the applicable requirement necessary?

A. It establishes what must be assessed and the acceptance basis.

Without the governing requirement, the ratio cannot be judged against the correct acceptance conditions.

3. The static factor is 2.0 and fatigue is outside the model. Which conclusion is justified?

B. Fatigue performance is not established by this model; it needs an appropriate assessment basis.

A static capacity-to-demand ratio is not a fatigue-life result. An excluded behavior is unassessed here, not automatically passed, failed, or irrelevant.

4. What should a compliance matrix connect?

C. Applicable requirements, evidence, and assessment status

The matrix supports traceability between the actual requirements, supporting records, and current conclusions.

06 Conversation answers

Safety Factors and Compliance

E Dialogue gaps

Pages 55-56

1. applicable requirement. Turn 2. The applicable requirement defines the governing obligation and acceptance basis missing from the current note.

2. factor of safety. Turn 4. The factor of safety is a defined ratio in this model, not a universal measure proving complete product safety.

3. failure criterion. Turn 6. The failure criterion defines the condition against which capacity and demand are assessed.

4. analysis limitations. Turn 8. Analysis limitations identify excluded behaviors, including fatigue and dynamic loading, that the reported model cannot establish.

5. model validity. Turn 10. Model validity concerns whether the model is appropriate for the claimed assessment, beyond simply producing a numerical output.

6. standard edition. Turn 12. The standard edition identifies the specific version whose requirements are being applied, rather than assuming all versions are interchangeable.

7. compliance matrix. Turn 14. A compliance matrix connects each applicable requirement with evidence and its assessment status.

8. conformity assessment. Turn 16. Conformity assessment evaluates specified requirements through the applicable route; writing an internal note does not establish its completion.

9. authorized sign-off. Turn 18. Authorized sign-off is an actual approval by the relevant authority, which is not supplied by the preliminary calculation.

10. evidence package. Turn 20. The evidence package must support the exact claim through identified requirements, suitable analyses, limitations, and review records.

F Transfer conversation

Page 57

1. static

Engineer: "The reported value belongs to the static model."

The briefing identifies the report as a preliminary static assessment.

2. fatigue

Reviewer: "The model excludes fatigue."

Fatigue is explicitly outside the scope of the supplied model.

3. unidentified

Engineer: "The acceptance threshold remains unidentified."

No applicable acceptance threshold is supplied in the briefing.

4. granted

Reviewer: "Release approval has not been granted."

The briefing states that the report has not received release approval.

07 Briefing and language checks

Field Issues and Customer Communication

A Briefing checks / page 58

1. What needs correction?

A. Humidity was stated as a confirmed cause without confirmation.

The earlier email converted an unconfirmed humidity hypothesis into a causal finding.

2. What can Marco commit to?

A. An update by 10 a.m. on October 2

The only supplied commitment is the next update, not resolution or operating authorization.

3. Which statement preserves the customer's report?

D. Unit L18 has two reported output interruptions.

The two reported events remain relevant even though their cause is unresolved.

D Language checks / page 63

1. Which opening best corrects the earlier message?

B. Our email overstated the cause; humidity remains unconfirmed.

The opening explicitly corrects the unsupported certainty without discarding the reported events.

2. Which deadline statement is supported?

A. I will send an investigation update by 10 a.m. on October 2.

Only the communication deadline is authorized by the supplied facts.

3. Which request could help distinguish causes without presuming humidity is responsible?

B. Provide event times, installed configuration, and available environmental records linked to L18.

Traceable observations let investigators test competing explanations. Filtering or labeling the evidence to fit the prior claim would prejudice the investigation.

4. What should the correction not imply?

B. Permission to change operation or perform an unapproved repair

The briefing gives no authorized operating or repair change, so the message must not create one.

07 Conversation answers

Field Issues and Customer Communication

E Dialogue gaps

Pages 64-65

1. causal statement. Turn 2. The causal statement claimed humidity produced the interruptions, which the available investigation has not established.

2. field report. Turn 4. The field report records the customer's two interruptions and remains relevant while the cause is unresolved.

3. working hypothesis. Turn 6. A working hypothesis remains under investigation without being presented as a supported causal finding.

4. correction notice. Turn 8. A correction notice explicitly amends the earlier claim so recipients do not continue relying on its unsupported certainty.

5. traceable record. Turn 10. A traceable record connects evidence to its source, event time, unit, and configuration rather than leaving context ambiguous.

6. resolution commitment. Turn 12. A resolution commitment promises an outcome or timing that remains unsupported while the cause and repair are unresolved.

7. interim update. Turn 14. An interim update reports progress before final resolution and should distinguish findings, uncertainties, and next steps.

8. service instruction. Turn 16. A service instruction must come through the authorized product-specific process; the correction does not authorize operating or repair changes.

9. case owner. Turn 18. The case owner coordinates the investigation communication and follow-up so the customer has a clear responsible contact.

10. Closure confirmation. Turn 20. Closure confirmation requires the actual case conditions to be satisfied; correcting an email alone does not resolve the underlying issue.

F Transfer conversation

Page 66

1. correction

Supplier: "Our earlier claim needs a correction."

The earlier message presented an unconfirmed explanation as an established cause.

2. hypothesis

Customer: "Contamination remains a hypothesis."

The briefing explicitly says contamination has not been confirmed as the cause.

3. update

Supplier: "Friday noon is the next update."

The promised milestone is a progress report rather than a completed repair.

4. open

Customer: "The investigation remains open."

Neither the cause nor the resolution has been established in the supplied facts.

08 Briefing and language checks

Systems Integration and Interface Control

A Briefing checks / page 67

1. What does 100 counts mean for H2?

D. 1 mm

H2 sends hundredths of a millimeter, so one hundred counts represent one millimeter.

2. What does S1 currently display for those 100 counts?

A. 100 mm

S1 still applies revision 1's interpretation of one millimeter for each transmitted count.

3. Which configuration has the earlier test evidence?

D. H1 with S1 under revision 1

The briefing identifies the earlier tests as H1, S1, and the approved revision-1 interface.

D Language checks / page 72

1. Which diagnosis matches the facts?

B. H2 and S1 use different scaling for the same transmitted count.

H2 encodes hundredths of a millimeter while S1 interprets whole millimeters.

2. For H2's 100-count message, which correction addresses the unit meaning?

C. Multiply 100 by 0.01 mm per count to obtain 1 mm; changing only the display label is insufficient.

Each count represents one hundredth of a millimeter. Multiplication by the encoded unit scale gives the intended physical value; a label change does not correct the number.

3. How should earlier evidence be treated?

A. Preserve it and assess which results remain applicable to the changed configuration.

Prior results remain valid records of the old configuration; reuse for the change needs justification.

4. What is needed before claiming integration readiness?

D. A resolved interface definition, controlled configuration, and appropriate reviewed test evidence

Readiness depends on a shared approved basis and evidence, not merely connectivity or preference.

08 Conversation answers

Systems Integration and Interface Control

E Dialogue gaps

Pages 73-74

1. unit scaling. Turn 2. Unit scaling converts the encoded count into a physical quantity; H2 now uses hundredths of a millimeter.

2. configuration mismatch. Turn 4. The configuration mismatch pairs hardware using the new scale with software expecting the old scale.

3. configuration record. Turn 6. The configuration record identifies the exact H1, S1, and revision-one combination behind the earlier evidence.

4. interface control document. Turn 8. The interface control document is the controlled shared definition; a draft revision does not replace the approved one automatically.

5. conversion factor. Turn 10. The conversion factor is 0.01 millimeters per count, converting H2's value of 100 into 1 millimeter.

6. change impact analysis. Turn 12. Change impact analysis identifies which functions, records, requirements, and tests are affected by the scaling change.

7. test evidence reuse. Turn 14. Test evidence reuse requires justified continued applicability; previous success cannot simply be transferred to the changed pairing.

8. regression tests. Turn 16. Regression tests check that the change has not broken previously working behavior in the defined affected scope.

9. interface owner. Turn 18. The interface owner maintains the shared definition and coordinates controlled changes across the participating teams.

10. integration readiness. Turn 20. Integration readiness requires the agreed configuration and supporting evidence, not merely understanding the source of a mismatch.

F Transfer conversation

Page 75

1. 5

Engineer: "The proposed distance is 5 mm."

Fifty counts multiplied by 0.1 millimeters per count equals five millimeters.

2. 50

Reviewer: "The old software displays 50 mm."

The old interpretation uses one millimeter per count, producing fifty millimeters.

3. scaling

Engineer: "The difference comes from unit scaling."

The same count is converted using two different physical-unit multipliers.

4. approval

Reviewer: "The proposed change still needs approval."

The briefing explicitly states that the new interface proposal is unapproved.

08

09 Conversation answers

A smoother plot is not a convergence study

E Extended conversation

Pages 76-77

1. quantity of interest. Turn 2. The quantity of interest is the specific output used for the intended assessment; different outputs may behave differently under refinement.

2. Mesh convergence. Turn 4. Mesh convergence concerns the behavior of a specified result as the discretization is refined; stability of one output does not establish it for every output.

3. stress singularity. Turn 6. A stress singularity can occur at an idealized geometric or loading feature, producing a local stress that does not converge to a finite value with refinement.

4. boundary conditions. Turn 10. Boundary conditions define how the modeled system is constrained or interacts with its surroundings; they must represent the intended assessment adequately.

5. Verification. Turn 14. Verification addresses whether the model implementation and numerical solution are correct for the specified model; it is distinct from assessing representation of reality.

6. sensitivity analysis. Turn 16. Sensitivity analysis examines how outputs change when selected inputs or assumptions vary; it helps identify which uncertainties matter to the conclusion.

F Transfer exchange

Page 78

1. peak

Lead: Stability of the average does not establish the local peak.

Different outputs may converge at different rates, so one stable measure cannot settle the other.

2. assessment

Analyst: State the specific quantity of interest for the assessment.

The intended assessment determines which output requires adequate supporting evidence.

3. report

Lead: Keep the remaining numerical limitation visible in the report.

The report must disclose the unresolved behavior instead of hiding it behind an average.

4. criterion

Analyst: A smooth image is not an acceptance criterion.

Visual appearance does not replace the defined technical basis for accepting the result.

09

Background reading: [COMSOL: Finite Element Mesh Refinement](#). The conversation is an original fictional teaching case.

10 Conversation answers

The replacement fits, but is it an approved alternate?

E Extended conversation

Pages 79-80

1. form-and-fit. Turn 2. Form-and-fit concerns physical characteristics and compatibility at the mounting or connection interface; it does not establish equivalent function or performance.

2. interface requirements. Turn 4. Interface requirements define the relevant mechanical, electrical, and information connections; a similar connector appearance does not demonstrate compliance.

3. transfer function. Turn 6. The transfer function here describes the relationship between the measured input and the sensor output; a changed scale can affect downstream interpretation.

4. bill of materials. Turn 12. The bill of materials identifies the parts belonging to the defined product configuration; listing a candidate as approved would misstate its status.

5. approved alternate. Turn 14. An approved alternate has completed the applicable assessment and approval for its defined use; a supplier recommendation or evaluation sample does not establish that status.

6. effectivity. Turn 16. Effectivity defines where or when an approved change applies, such as specified builds or serial ranges, avoiding an uncontrolled mix of configurations.

F Transfer exchange

Page 81

1. use

Engineer: These parts were ordered for evaluation, not production use.

The order's stated purpose is assessment and does not authorize installation in production products.

2. history

Buyer: Correct the status without losing the purchase history.

The record should accurately show the correction while preserving what was ordered and why.

3. requirements

Engineer: Complete the comparison against the applicable requirements.

Equivalence must be assessed against the relevant product requirements, not assumed from availability.

4. alternates

Buyer: Keep the candidate separate from approved alternates.

An unassessed candidate must not appear to have the same status as an authorized substitute.

11 Conversation answers

Two settings changed, so which one improved the result?

E Extended conversation

Pages 82-83

1. factors. Turn 2. Factors are the input variables being studied; both fan setting and vent design changed between these runs, so their separate effects are not established.

2. design matrix. Turn 4. The design matrix identifies the planned combinations of factor settings; it exposes missing combinations needed for the intended comparison.

3. interaction. Turn 6. An interaction means the effect of one factor depends on the level of another; separate single-change conclusions may miss that relationship.

4. response variable. Turn 8. The response variable is the measured outcome of the experiment, with its location, timing, and calculation defined so results are comparable.

5. replicates. Turn 10. Replicates repeat the experimental treatment on appropriate independent experimental units or runs; multiple readings within one run do not automatically provide them.

6. randomization. Turn 14. Randomization varies the allocation or order to reduce systematic association with uncontrolled effects; practical restrictions must be handled explicitly in the design.

F Transfer exchange

Page 84

1. trials

Lead: Ten readings from one run are not automatically ten independent trials.

Measurements within a single run do not automatically represent independent repetitions of the experiment.

2. variation

Engineer: Identify the experimental unit and the source of variation.

The design must establish what is independently repeated and which variation the evidence captures.

3. count

Lead: Keep the measurement count distinct from the run count.

The number of readings and the number of test runs answer different questions.

4. supports

Engineer: Limit the conclusion to what the design actually supports.

A conclusion must reflect the experimental design rather than an inflated apparent sample size.

11

Background reading: [NIST: Choosing an Experimental Design](#). The conversation is an original fictional teaching case.

Find the move you need

Use these as precise language models. Replace details only when you know the actual facts, authority, and agreed conditions.

01 Requirements and Constraints

Clarify the adjective: What maximum mass would count as lightweight?

State the proposal: We propose a maximum mass of 1.2 kg.

Identify a missing input: The handling conditions have not been specified.

Full phrase set: pages 7-8 | Dialogue: pages 10-11

02 Design Reviews and Technical Pushback

Locate the concern: My concern is the evidence for R-14.

Ask for the analysis: Where is the documented tolerance stack-up?

Bound the action: Confirm the revision and attach the stack-up for this interface.

Full phrase set: pages 16-17 | Dialogue: pages 19-20

03 Failure Modes and Reliability

State the observation: S7 stopped producing an output at minute 46.

Qualify the explanation: Connector damage is one hypothesis under investigation.

Request discriminating evidence: Which observation would distinguish the competing explanations?

Full phrase set: pages 25-26 | Dialogue: pages 28-29

04 Testing, Validation, and Data Interpretation

Lead with the result: All three prototypes met the stated criterion.

State the criterion: The response-time limit was 200 milliseconds.

Avoid population overclaim: That is not proof that every production unit will pass.

Full phrase set: pages 34-35 | Dialogue: pages 37-38

Keep the conversation productive

Use these as precise language models. Replace details only when you know the actual facts, authority, and agreed conditions.

05 Manufacturability and Cost Engineering

Set the comparison: Use the same 3,000-unit quantity for both options.

Add recurring costs: A has \$14 of material and assembly cost per unit.

Explain the difference: B is \$2,000 lower at this volume.

Full phrase set: pages 43-44 | Dialogue: pages 46-47

06 Safety Factors and Compliance

Ask for the basis: Compliant with which applicable requirement?

Limit the result: This is a preliminary static-model result.

Separate analysis from approval: A calculated result is not authorized sign-off.

Full phrase set: pages 52-53 | Dialogue: pages 55-56

07 Field Issues and Customer Communication

Acknowledge the correction: Our earlier email overstated the cause.

Recognize the report: We have recorded two interruptions on unit L18.

Separate the commitments: I can commit to an update, not a resolution date.

Full phrase set: pages 61-62 | Dialogue: pages 64-65

08 Systems Integration and Interface Control

Identify the pairing: Which hardware build and software version are under test?

Show the consequence: One hundred counts means 1 mm for H2 but 100 mm for S1.

Avoid blanket reuse: That result does not automatically cover the new pairing.

Full phrase set: pages 70-71 | Dialogue: pages 73-74

Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

acceptance criterion	agree an acceptance criterion	5
acceptance limit	compare with the acceptance limit	32
action closure	verify action closure	15
affected test	identify affected tests	69
allowable	identify the allowable	50
ambiguity	resolve ambiguity	6
analysis limitation	state analysis limitations	51
anomaly	record an anomaly	23
applicable requirement	identify applicable requirements	51
assembly cost	reduce assembly cost	41
assumption log	review the assumption log	15
assumption register	update the assumption register	6
authorized sign-off	obtain authorized sign-off	51
backward compatibility	assess backward compatibility	69
baseline	establish a baseline	6
bill of materials	compare the bill of materials	41
boundary condition	examine boundary conditions	32
break-even volume	calculate break-even volume	42
buckling	assess buckling	50
byte order	specify byte order	68
CAD model	update the CAD model	14
case owner	name the case owner	60
case reference	use the case reference	59
causal statement	correct a causal statement	59

Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

censored observation	account for censored observations	33
chain of custody	maintain chain of custody	60
change approval	obtain change approval	69
change impact analysis	complete change impact analysis	69
clearance	verify minimum clearance	14
closure confirmation	obtain closure confirmation	60
closure evidence	attach closure evidence	15
compliance matrix	maintain a compliance matrix	51
concurrence	obtain reviewer concurrence	15
conditional approval	document conditional approval	15
confidence interval	report a confidence interval	33
configuration baseline	establish the configuration baseline	68
configuration mismatch	identify a configuration mismatch	69
configuration record	preserve the configuration record	69
confirmed finding	separate confirmed findings	59
conformity assessment	identify the conformity assessment route	51
containment	define containment	24
conversion factor	apply the conversion factor	69
correction notice	issue a correction notice	59
corrective action	evaluate corrective action	24
cost driver	identify the cost driver	41
cost model	review the cost model	41
cost trade-off	explain the cost trade-off	42
customer impact	acknowledge customer impact	59

Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

cycle time measure cycle time	42
data type specify the data type	68
datum identify the datum	14
derived requirement justify a derived requirement	6
design constraint work within a design constraint	5
design for assembly apply design for assembly	41
design for manufacture review design for manufacture	41
design freeze prepare for design freeze	15
design load define the design load	50
design margin preserve design margin	6
design requirement define a design requirement	5
design review conduct a design review	14
detectability assess detectability	24
deviation approval record deviation approval	51
diagnostic evidence preserve diagnostic evidence	24
dimensional tolerance apply dimensional tolerances	14
dissent document unresolved dissent	15
drawing revision confirm the drawing revision	14
duty cycle specify the duty cycle	5
dynamic loading assess dynamic loading	50
environmental condition document environmental conditions	5
escalation contact identify the escalation contact	60
event log export the event log	24
evidence boundary state the evidence boundary	33

Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

evidence package	assemble the evidence package	51
evidence request	make an evidence request	59
extrapolation	avoid unsupported extrapolation	33
factor of safety	state the factor of safety	50
failure analysis	commission failure analysis	23
failure criterion	specify the failure criterion	50
failure effect	assess the failure effect	23
failure mechanism	investigate the failure mechanism	23
failure mode	describe the failure mode	23
fatigue	evaluate fatigue	50
fault isolation	support fault isolation	24
field action	assess a field action	60
field report	assess a field report	59
field width	confirm the field width	68
FMEA	update the FMEA	24
hardware build	identify the hardware build	68
hypothesis	test a hypothesis	23
impact statement	prepare an impact statement	60
incremental cost	compare incremental cost	42
independent review	arrange independent review	51
installed configuration	document the installed configuration	59
instrument calibration	check instrument calibration	32
integration readiness	assess integration readiness	69
integration test	run an integration test	68

Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

interface contract	define the interface contract	68
interface control document	update the interface control document	68
interface owner	identify the interface owner	69
interference	identify potential interference	14
interim update	provide an interim update	60
intermittent fault	investigate an intermittent fault	23
load case	assess the load case	50
load combination	document the load combination	50
make-or-buy decision	support a make-or-buy decision	42
manufacturability	assess manufacturability	41
mass budget	allocate the mass budget	5
material cost	estimate material cost	41
mating surface	inspect the mating surface	14
maximum mass	specify maximum mass	5
measurement uncertainty	evaluate measurement uncertainty	32
message schema	version the message schema	69
model validity	assess model validity	51
nominal dimension	check the nominal dimension	14
nonrecurring cost	identify nonrecurring cost	41
normative reference	check normative references	51
objective value	distinguish threshold and objective	6
observed maximum	report the observed maximum	32
occurrence	estimate occurrence	24
open item	assign an open item	15

Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

operating envelope	define the operating envelope	5
operating history	request operating history	59
part consolidation	evaluate part consolidation	42
pass fraction	calculate the observed pass fraction	33
payload	inspect the payload	68
population	define the target population	33
process capability	assess process capability	42
production volume	state production volume	42
rationale	record the rationale	6
recurrence	monitor recurrence	24
recurring cost	compare recurring cost	41
regression test	define regression tests	68
release authority	identify the release authority	15
release recommendation	qualify a release recommendation	15
release restriction	communicate release restrictions	51
reliability claim	qualify a reliability claim	33
repeatability	check repeatability	23
reported symptom	record the reported symptom	59
representativeness	assess representativeness	33
reproducibility	assess reproducibility	23
requirement allocation	review requirement allocation	6
requirement owner	identify the requirement owner	6
residual uncertainty	state residual uncertainty	24
resolution	specify instrument resolution	32

Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

resolution commitment	avoid an unsupported resolution commitment	60
response time	measure response time	32
return authorization	confirm return authorization	60
review action	record a review action	15
review package	complete the review package	14
root cause	establish the root cause	23
sample size	report the sample size	32
scrap allowance	include a scrap allowance	42
service instruction	follow the service instruction	60
service life	state the service life	5
severity	assess severity	24
software version	record the software version	68
specimen	preserve the specimen	24
stakeholder need	capture the stakeholder need	5
standard edition	confirm the standard edition	51
static analysis	review static analysis	50
system boundary	define the system boundary	5
takt time	compare cycle time with takt time	42
technical objection	state a technical objection	15
test article	identify the test article	32
test condition	record test conditions	32
test coverage	identify gaps in test coverage	32
test deviation	assess a test deviation	33
test duration	state the test duration	33

Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

test evidence reuse justify test evidence reuse	69
test fixture inspect the test fixture	23
test protocol approve the test protocol	32
test setup verify the test setup	23
testability improve testability	6
threshold specify the threshold	6
tolerance relaxation assess tolerance relaxation	42
tolerance stack-up calculate the tolerance stack-up	14
tooling investment justify tooling investment	41
traceability maintain traceability	6
traceable record retain a traceable record	60
ultimate capacity state ultimate capacity	50
unit cost calculate unit cost	41
unit scaling document unit scaling	68
unit serial number confirm the unit serial number	59
update cadence agree an update cadence	60
validation plan intended-use validation	33
verification complete requirement verification	33
verification method select a verification method	5
version negotiation define version negotiation	69
working hypothesis qualify the working hypothesis	59
worst-case analysis document a worst-case analysis	14
yield assumption challenge the yield assumption	42
yield limit identify the yield limit	50

Original cases. Clear boundaries.

These are original fictional teaching conversations, not transcripts of real workers. Names, figures, organizations, and incidents are invented. Sources support terminology and communication aims, not the fictional results or an endorsement of this book.

[1] NASA. Systems Engineering Handbook, Appendix C: How to Write a Good Requirement.

[Open the original source](#)

Background on clear, verifiable requirements. The enclosure requirements and customer discussion are original fictional examples. Checked 10 October 2026.

[2] NASA. Systems Engineering Handbook, Section 6: Crosscutting Technical Management.

[Open the original source](#)

Background on interface and configuration management. The book does not prescribe NASA procedures for other organizations. Checked 10 October 2026.

[3] ASQ. What Is Failure Mode and Effects Analysis?

[Open the original source](#)

Background on distinguishing failure modes, effects, and possible causes. No scoring scale or acceptance rule is asserted as universal. Checked 10 October 2026.

[4] NIST/SEMATECH. Engineering Statistics Handbook, Section 8.1.3.2: Lack of Failures.

[Open the original source](#)

Background on the limits of reliability information when few failures are observed. The prototype case is language practice, not a statistical qualification plan. Checked 10 October 2026.

Phil Stilwell

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